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MIL-M-28787/353(SH)
20 February 1990

MILITARY SPECIFICATION

MODULES, DIGITAL, STANDARD ELECTRONIC

MICROPROCESSOR, 68000 MODULE, KEY CODE NAH

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the modules described herein shall consist of this specification and the latest issue of MIL-M-28787.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic 68000 microprocessor module. This module is class I as specified in MIL-STD-1389.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, Code SEA-55Z3, Department of the Navy, Washington, D.C. 20362-5101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification, standards, and handbook. The following specification, standards, and handbook form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

HANDBOOK

MILITARY

DOD-HDBK-217 - Reliability Prediction of Electronic Equipment

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Qualification. Modules furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list by the time of award of contract (see 4.5).

3.2 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.3 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.4 Failure rate requirements. All modules procured to this specification shall have a failure rate predicted in accordance with MIL-STD-1389. The predicted failure rate shall not exceed 1.6970 failures per million operating hours when predicted in accordance with DOD-HDBK-217E.

3.5 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.6 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.6.1 Key code. The module key code shall be NAH.

3.6.2 Size. The module size shall be 2A Format B.

3.6.3 Weight. The module shall have a maximum weight of 120 grams.

3.6.4 Contact pins. The module shall have 100 contact pins positioned 1 through 50 in one row and 51 through 100 in the other row.

3.6.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.6.6 Depth. The module shall have no additional depth requirements.

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TABLE I. Module electrical requirements.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		0 ± 5°C 4/		60 ± 5°C 4/		Units
				Min	Max	Min	Max	Min	Max	
A	Isolation	II	S	10	INF	10	INF	10	INF	MOhms
B1-B6	Continuity	III	X	0.0	1.0	0.0	1.0	0.0	1.0	Ohm
B7-B12	Continuity	III	Q	0.0	0.1	0.0	0.1	0.0	0.1	Ohm
C	Vcc stress	IV	Q	----------	Module shall pass all			remaining tests	-------	
D1-D27	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D28-D30	Ibd	V	S	0.0	0.12	0.0	0.12	0.0	0.12	mA
D31-D38	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0.2	mA
D39-D43	Ibd	V	S	0.0	1.0	0.0	1.0	0.0	1.0	mA
E1-E42	Vic	VI	S	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	V
E43	Vic	VI	S	-1.2	-0.4	-1.2	-0.4	-1.2	-0.4	V
F1-F16	Ios	VII	S	-225.0	-40.0	-225.0	-40.0	-225.0	-40.0	mA
F17-F19	Ios	VII	S	-100.0	-40.0	-100.0	-40.0	-100.0	-40.0	mA
F20-F27	Ios	VII	S	-130.0	-6.0	-130.0	-6.0	-130.0	-6.0	mA
F28	Ios	VII	S	-100.0	-15.0	-100.0	-15.0	-100.0	-15.0	mA
F29	Ios	VII	S	-250.0	-30.0	-250.0	-30.0	-250.0	-30.0	mA
F30	Ios	VII	S	-130.4	-30.0	-130.4	-30.0	-130.4	-30.0	mA
F31,F32	Ios	VII	S	-552.6	-500.0	-552.6	-500.0	-552.6	-500.0	mA
F33-F35	Ios	VII	S	-130.95	-30.6	-130.95	-30.6	-130.95	-30.6	mA
G1-G16	Iih	VIII	X	-20	20	-20	20	-20	20	uA
G17-G21	Iih	VIII	X	0	20	0	20	0	20	uA
G22-G28	Iih	VIII	X	-268	-203	-268	-203	-268	-203	uA
G29-G31	Iih	VIII	X	-288	-203	-288	-203	-288	-203	uA
G32-G38	Iih	VIII	X	-269	-223	-269	-223	-269	-223	uA
G39	Iih	VIII	X	-268	-193	-268	-193	-268	-193	uA
G40-G42	Iih	VIII	X	-300	-246	-300	-246	-300	-246	uA
G43	Iih	VIII	X	0	25	0	25	0	25	uA
H1-H16	Iil	IX	X	-200	0	-200	0	-200	0	uA
H17	Iil	IX	X	-250	0	-250	0	-250	0	uA
H18-H22	Iil	IX	X	-380	-1	-380	-1	-380	-1	uA

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		0 ± 5°C 4/		60 ± 5°C 4/		Units
				Min	Max	Min	Max	Min	Max	
H23-H29	Iil	IX	X	-1270	-722	-1270	-722	-1270	-722	uA
H30,H31	Iil	IX	X	-891	-492	-891	-492	-891	-492	uA
H32	Iil	IX	X	-891	-463	-891	-463	-891	-463	uA
H33-H35	Iil	IX	X	-761	-462	-761	-462	-761	-462	uA
H36-H38	Iil	IX	X	-931	-542	-931	-542	-931	-542	uA
H39-H41	Iil	IX	X	-911	-562	-911	-562	-911	-562	uA
H42	Iil	IX	X	-871	-582	-871	-582	-871	-582	uA
H43	Iil	IX	X	-2500	-1452	-2500	-1452	-2500	-1452	uA
J1-J53	Voh	X	X	2.4	4.75	2.4	4.75	2.4	4.75	V
J54-J68	Voh	X	X	2.5	4.75	2.5	4.75	2.5	4.75	V
K1-K30	Vol	XI	X	0.0	0.4	0.0	0.4	0.0	0.4	V
K31-K68	Vol	XI	X	0.0	0.5	0.0	0.5	0.0	0.5	V
L1-L42	Iozh	XII	X	-20	20	-20	20	-20	20	uA
L43	Iozh	XII	X	-20	60	-20	60	-20	60	uA
L44-L46	Iozh	XII	X	-20	45	-20	45	-20	45	uA
L47	Iozh	XII	X	-346	-226	-346	-226	-346	-226	uA
M1-M16	Iozl	XIII	X	-200	0	-200	0	-200	0	uA
M17-M42	Iozl	XIII	X	-20	20	-20	20	-20	20	uA
M43	Iozl	XIII	X	-420	20	-420	20	-420	20	uA
M44-M46	Iozl	XIII	X	-270	20	-270	20	-270	20	uA
M47	Iozl	XIII	X	-770	-466	-770	-466	-770	-466	uA
N	Icc	XIV	X	36.33	726.64	36.33	726.64	36.33	726.64	mA
P1,P2	Functional	XV	X	----------See table XV----------						
Q1,Q2	Dynamic	XVI	S	----------See table XVI----------						
Q3,Q4	Dynamic	XVI	Q	----------See table XVI----------						
R	PSFC	XVIII	S	3.78	7.02	3.78	7.02	3.78	7.02	uF

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		0 ± 5°C 4/		60 ± 5°C 4/		Units
				Min	Max	Min	Max	Min	Max	
S	PSTA	XIX	Q	0	400	0	400	0	400	mV
T1-T64	Output crosstalk, high level	XX	Q	0	400	0	400	0	400	mV
T65-T67	Output crosstalk, high level	XX	Q	0	500	0	500	0	500	mV
T68	Output crosstalk, high level	XX	Q	0	400	0	400	0	400	mV
U1-U64	Output crosstalk, low level	XXI	Q	0	400	0	400	0	400	mV
U65-U67	Output crosstalk, low level	XXI	Q	0	500	0	500	0	500	mV
U68	Output crosstalk, low level	XXI	Q	0	700	0	700	0	700	mV
V	Output crosstalk, duration	XXII	Q	0	50	0	50	0	50	nS
W1-W16	Cin	XXIII	Q	2	35	2	35	2	35	pF
W17-W19	Cin	XXIII	Q	2	20	2	20	2	20	pF
W20-W25	Cin	XXIII	Q	2	25	2	25	2	25	pF
W26-W31	Cin	XXIII	Q	2	40	2	40	2	40	pF
W32,W33	Cin	XXIII	Q	2	15	2	15	2	15	pF
W34-W36	Cin	XXIII	Q	2	55	2	55	2	55	pF
W37-W43	Cin	XXIII	Q	2	30	2	30	2	30	pF

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		0 ± 5°C 4/		60 ± 5°C 4/		Units
				Min	Max	Min	Max	Min	Max	
X1-X27	Cout	XXIV	Q	2	25	2	25	2	25	pF
X28-X30	Cout	XXIV	Q	2	40	2	40	2	40	pF
X31	Cout	XXIV	Q	2	55	2	55	2	55	pF
Y1-Y23	Disable time	XXV	Q	2.0	161.5	2.0	161.5	2.0	161.5	nS
Y24-Y46	Disable time	XXV	Q	2.0	246.0	2.0	246.0	2.0	246.0	nS
Y47,Y48	Disable time	XXV	Q	2.0	183.5	2.0	183.5	2.0	183.5	nS
Y49	Disable time	XXV	Q	2.0	158.5	2.0	158.5	2.0	158.5	nS
Y50-Y81	Disable time	XXV	Q	7.0	187.5	7.0	187.5	7.0	187.5	nS
Z	Threshold	XXVI	X	See table XXVI						
AA1-AA23	Noise margin	XXVII	Q	0	900	0	900	0	900	mV

- See footnotes on next page.

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1/ "X" parameters shall be 100 percent tested during the module acceptance test.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified in the test table, limits are for initial and end-of-life.

5/ Abbreviations used in table I are defined in 6.1.

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3.6.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.6.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part or identifying number M28787/353-1. No marking shall interfere with heat dissipation from the fin (such as affixing a label which may act as a heat barrier).

3.7 Circuit pin configuration. The functional block diagram shall conform to figure 1. The pin assignments shall conform to figure 2.

3.8 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 in table I.

3.9 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits in table I (see 6.2).

3.10 Power supply voltage. The module shall be capable of conforming to the requirements in table I for a power supply voltage (Vcc) of 5.0 $\pm$ 0.5 V dc, except the output short circuit current limits apply only for a power supply voltage of 5.5 V dc $\pm$ 1 percent.

3.11 Parallel outputs. Paralleling of outputs is not permissible.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. Unless otherwise specified herein, all resistors shall be metal film and all capacitors shall be ceramic.

4.4 Test procedures. The module shall be tested as specified in table I.

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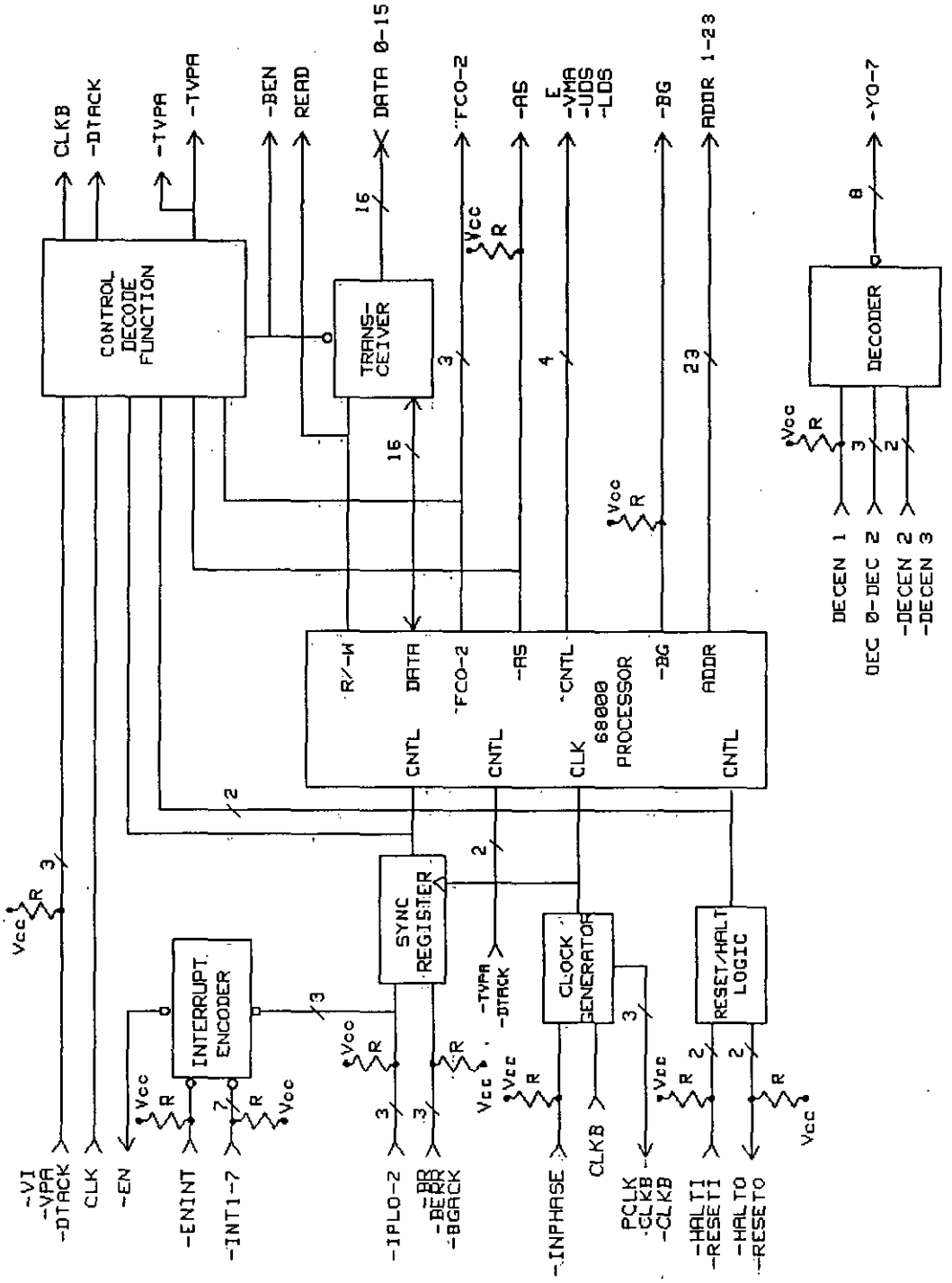


FIGURE 1. NAH functional block diagram.

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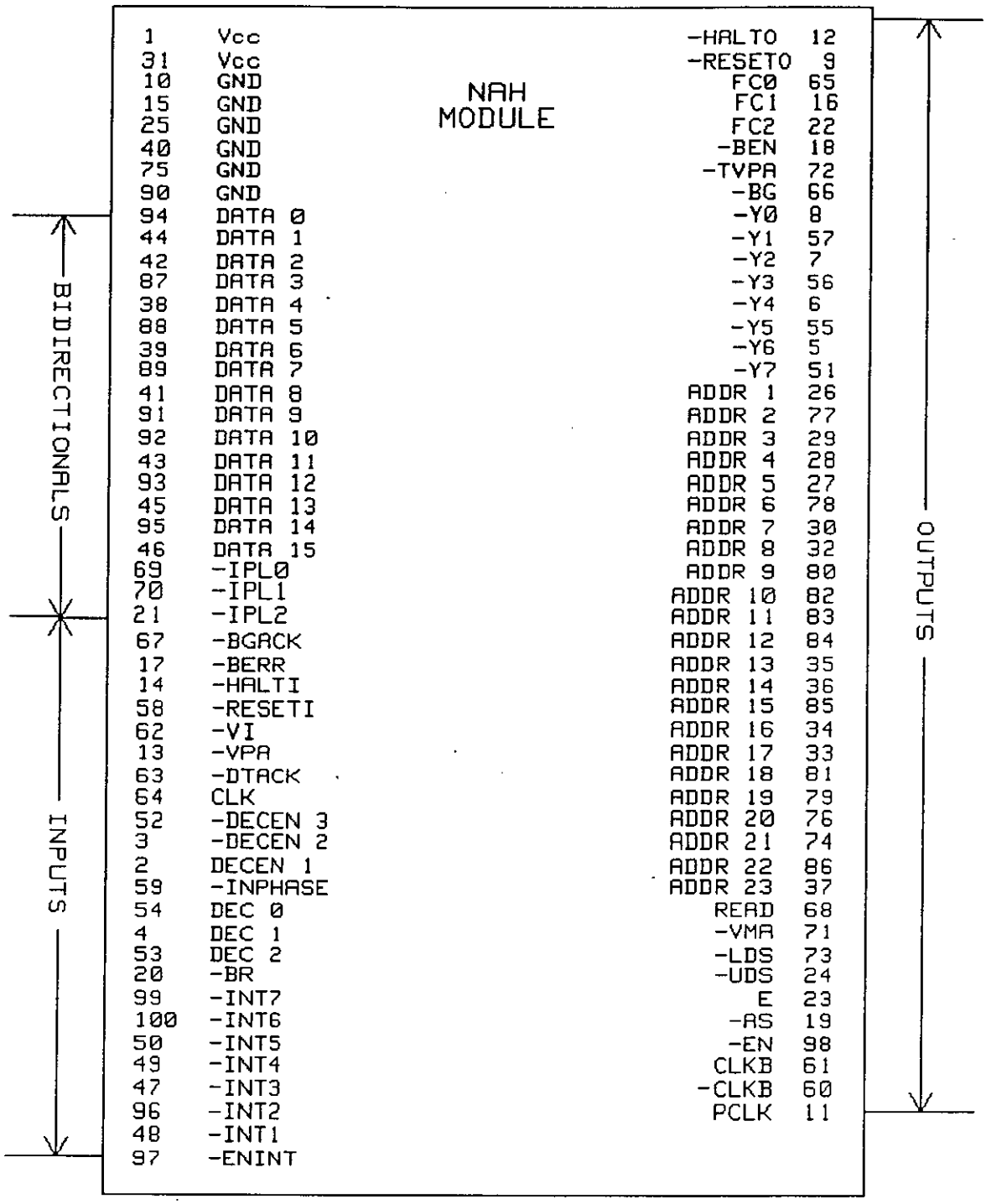


FIGURE 2. NAH pin assignment.

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4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Quality Assurance Activity (SEM-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through AA shall be performed in any order with the following exceptions. Test conditions A, C, D, E and F are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, C, D, E or F are part of a sequence of tests, the order of testing shall be as follows: A, C, D, E and F where applicable, followed by the remaining tests in the sequence.

4.4.3 Multiple testing of pins. Singular pins listed multiple times in a test table indicate multiple devices connected to the pins. The module shall be tested so that each device connected to a pin is exercised and tested.

4.5 Qualification testing. Initial qualification tests and periodic check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through AA in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B, G through V, Y, Z and AA shall be performed in the order specified in 4.4.2 using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 3. The input waveforms are shown on figure 4.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figures 5 and 6.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 7. Following the vibration tests, the electrical tests in table I which are marked with an "X" in the test type column, and test condition R, shall be performed using the initial limits at 25 ± 5°C ambient.

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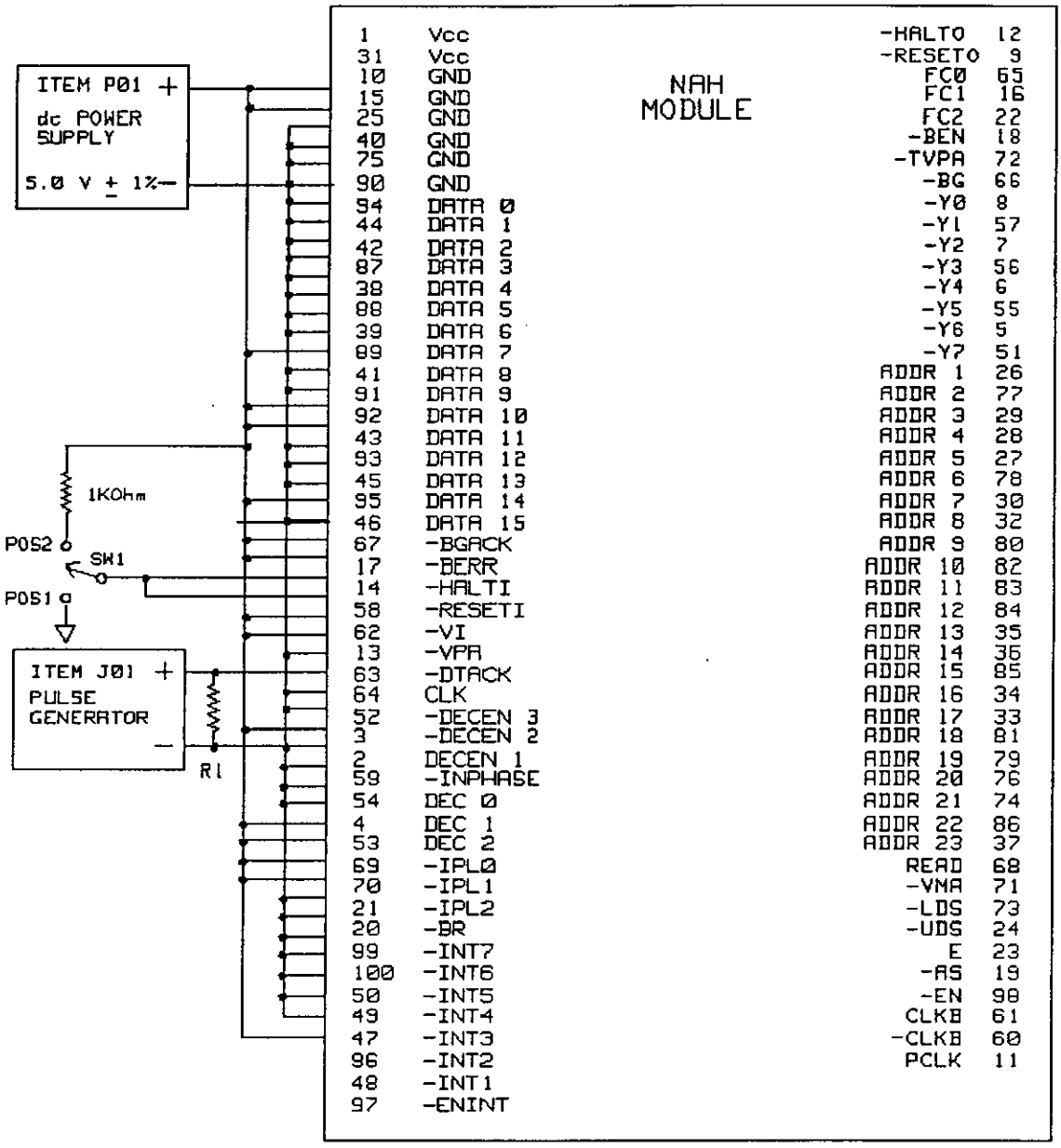


FIGURE 3. NAH operating temperature and life test circuit.

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NOTES:

1. The load for each output is as specified in the output voltage test.

2. R1 = 51 Ohms ± 5 percent, 1 W.

3. Apply the dc power (item P01) before applying the pulse generator (item J01).

4. The setup for item J01 when connected to R1 and not connected to pin 64 is shown on figure 4.

5. Connect item J01 to pin 64, then set SW1 to POS 1 for 100 mS (minimum) and then set it to POS 2.

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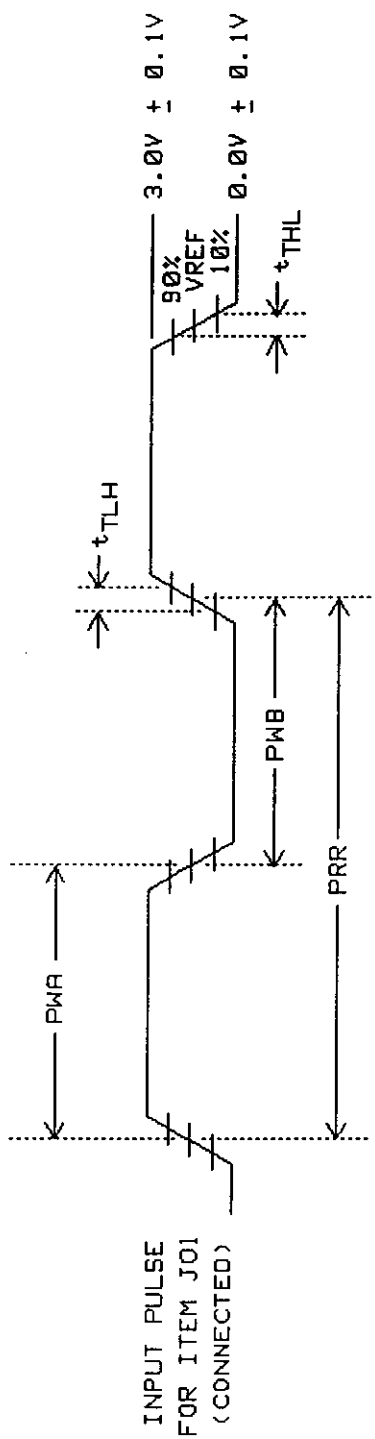


FIGURE 4. Operating temperature and life test waveforms.

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NOTE:

1. Input pulse parameters for item J01:

   PWA = 125 ± 25 nS.
   PWB = 125 ± 25 nS.
   PRR = 250 ± 50 nS.
   tTLH = tTHL = 10 ± 1 nS.

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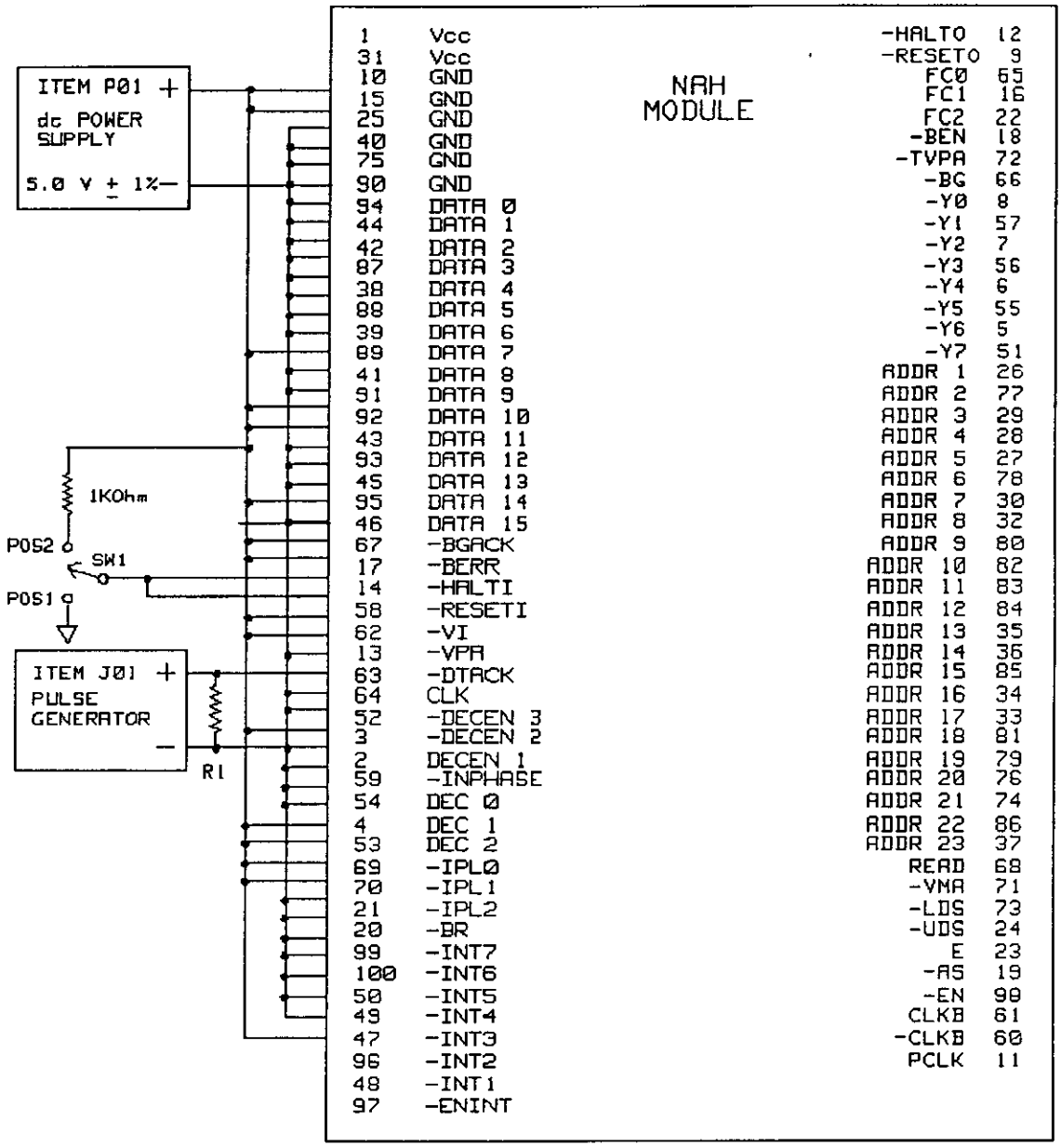


FIGURE 5. NAH component temperature test circuit.

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NOTES:

1. The load for each output is as specified in the output voltage test.

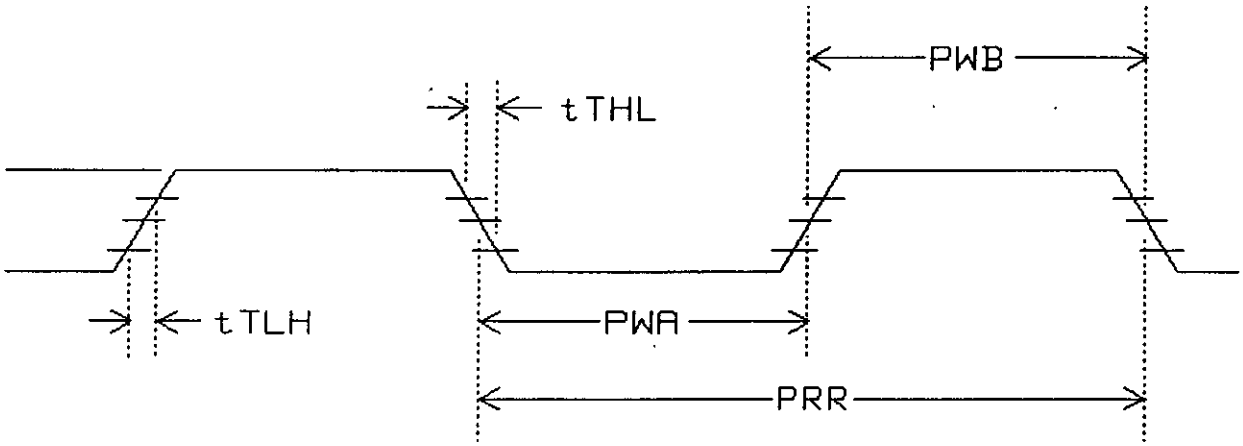
2. R1 = 51 Ohms ± 5 percent, 1 W.

3. Apply the dc power (item P01) before applying the pulse generator (item J01).

4. The setup for item J01 when connected to R1 and not connected to pin 64 is shown on figure 4.

5. Connect item J01 to pin 64, then set SW1 to POS 1 for 100 mS (minimum) and then set it to POS 2.

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NOTE:

1. Input pulse parameters for item J01:

   PWA = 125 ± 25 nS.
   PWB = 125 ± 25 nS.
   PRR = 250 ± 50 nS.
   tTLH = tTHL = 10 ± 1 nS.

FIGURE 6. Component temperature test waveforms.

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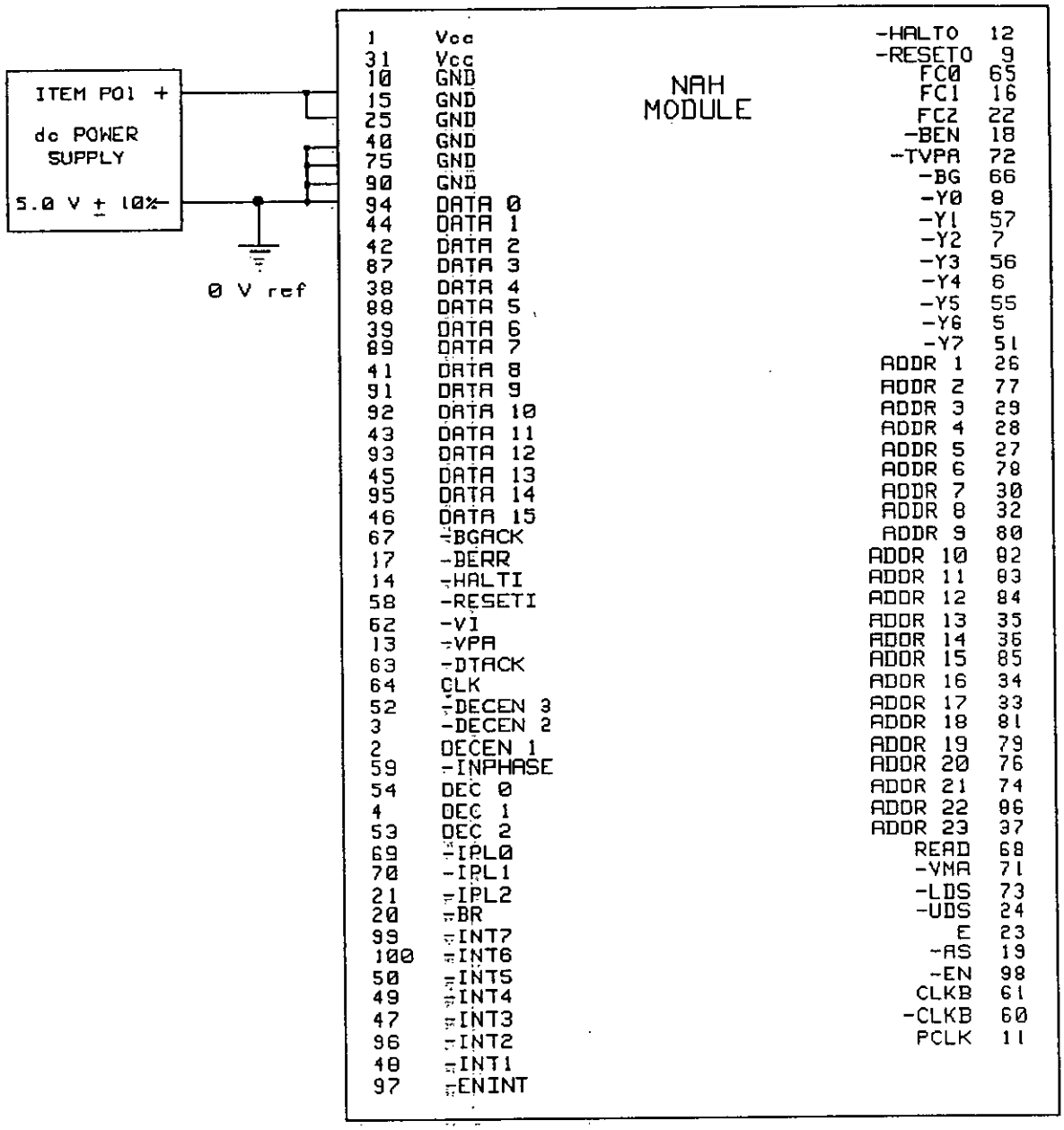


FIGURE 7. NAH vibration test circuit.

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4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figures 3 and 4. Following the life test and while still at the maximum class temperature, electrical tests B, G through V, Y, Z and AA shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements in table I. Following the return of the module to 25 ± 5°C, electrical tests A, B, G through V, Y, Z and AA shall be performed in the order specified in 4.4.2 using the 25 ± 5°C ambient end-of-life requirements in table I.

4.5.4 Thermal shock test. The thermal shock test shall be conducted in accordance with MIL-M-28787. The module shall be unpowered during this test. Following the thermal shock test, the electrical tests in table I which are marked with an "X" in the test type column, shall be performed using the initial limits at 25 ± 5°C ambient.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Definitions. Negative (-) current is current directed out of the module pin and positive (+) current is current directed into the module pin. Abbreviations used in this specification are defined as follows:

(a) Cin = input capacitance.
(b) Cout = output capacitance.
(c) Ibd = input breakdown current.
(d) Icc = power supply current.
(e) Iih = input high level current.
(f) Iil = input low level current.
(g) Iin = input forcing current.
(h) Ioh = output high level current.
(i) Iol = output low level current.
(j) Ios = output short circuit current.
(k) Iozh = output high level leakage current.
(l) Iozl = output low level leakage current.
(m) NA = not applicable.
(n) PSFC = power supply filter capacitance.
(o) PSTA = power supply transient amplitude.
(p) Vic = input clamp diode voltage.
(q) Vih = input high level voltage.
(r) Vil = input low level voltage.
(s) Vin = input forcing voltage.
(t) Voh = output high level voltage.
(u) Vol = output low level voltage.
(v) Vout = output forcing voltage.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements in table I, the module is considered to have completed its useful life (see 3.9).

6.3 Module description. The following electrical parameters reflect the 0°C to 60°C end-of-life parameters in table I. Operating power supply voltage range: 4.5 V dc to 5.5 V dc. Module power dissipation: 3.815 W (max); 2.327 W (typical).

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6.3.1 Input parameters.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (uA)	Iih max (uA)	Ibd max (uA)	Cin max (pF)
DATA 0	-1.5	0.7	2.0	-200	20	100	35
DATA 1	-1.5	0.7	2.0	-200	20	100	35
DATA 2	-1.5	0.7	2.0	-200	20	100	35
DATA 3	-1.5	0.7	2.0	-200	20	100	35
DATA 4	-1.5	0.7	2.0	-200	20	100	35
DATA 5	-1.5	0.7	2.0	-200	20	100	35
DATA 6	-1.5	0.7	2.0	-200	20	100	35
DATA 7	-1.5	0.7	2.0	-200	20	100	35
DATA 8	-1.5	0.7	2.0	-200	20	100	35
DATA 9	-1.5	0.7	2.0	-200	20	100	35
DATA 10	-1.5	0.7	2.0	-200	20	100	35
DATA 11	-1.5	0.7	2.0	-200	20	100	35
DATA 12	-1.5	0.7	2.0	-200	20	100	35
DATA 13	-1.5	0.7	2.0	-200	20	100	35

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (uA)	Iih max (uA)	Ibd max (uA)	Cin max (pF)
DATA 14	-1.5	0.7	2.0	-200	20	100	35
DATA 15	-1.5	0.7	2.0	-200	20	100	35
-BGACK	-1.5	0.7	2.0	-911	-268	100	40
-BERR	-1.5	0.7	2.0	-911	-268	100	40
-BR	-1.5	0.7	2.0	-911	-268	100	40
-HALTI	-1.5	0.7	2.0	-891	-268	100	25
-RESETI	-1.5	0.7	2.0	-891	-268	100	25
-VI	-1.5	0.7	2.0	-761	-300	1000	40
-VPA	-1.5	0.7	2.0	-761	-300	1000	40
-DTACK	-1.5	0.7	2.0	-761	-300	1000	40
DEC 0	-1.5	0.7	2.0	-380	20	100	20
DEC 1	-1.5	0.7	2.0	-380	20	100	20
DEC 2	-1.5	0.7	2.0	-380	20	100	20
-DECEN 3	-1.5	0.7	2.0	-380	20	100	15

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (uA)	Iih max (uA)	Ibd max (uA)	Cin max (pF)
-DECEN 2	-1.5	0.7	2.0	-380	20	100	15
-DECEN 1	-1.5	0.7	2.0	-891	-268	100	25
-INPHASE	-1.2	0.8	2.0	-2500	-268	1000	25
-IPL0	-1.5	0.7	2.0	-931	-288	120	55
-IPL1	-1.5	0.7	2.0	-931	-288	120	55
-IPL2	-1.5	0.7	2.0	-931	-288	120	55
-INT1	-1.5	0.7	2.0	-1270	-268	200	30
-INT2	-1.5	0.7	2.0	-1270	-268	200	30
-INT3	-1.5	0.7	2.0	-1270	-268	200	30
-INT4	-1.5	0.7	2.0	-1270	-268	200	30
-INT5	-1.5	0.7	2.0	-1270	-268	200	30
-INT6	-1.5	0.7	2.0	-1270	-268	200	30
-INT7	-1.5	0.7	2.0	-1270	-268	200	30

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (uA)	Iih max (uA)	Ibd max (uA)	Cin max (pF)
-ENINT	-1.5	0.7	2.0	-871	-268	200	25
CLK	-1.5	0.7	2.0	-250	25	1000	25

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6.3.2 Output parameters.

Signal	Voh min (V)	Vol max (V)	Ioh min (uA)	Iol max (mA)	Iozh max (uA)	Iozl min (uA)	Ios max (mA)	Cout max (pF)
DATA 0	2.4	0.4	-3000	12.0	20	-200	-40.0	35
DATA 1	2.4	0.4	-3000	12.0	20	-200	-40.0	35
DATA 2	2.4	0.4	-3000	12.0	20	-200	-40.0	35
DATA 3	2.4	0.4	-3000	12.0	20	-200	-40.0	35
DATA 4	2.4	0.4	-3000	12.0	20	-200	-40.0	35
DATA 5	2.4	0.4	-3000	12.0	20	-200	-40.0	35
DATA 6	2.4	0.4	-3000	12.0	20	-200	-40.0	35
DATA 7	2.4	0.4	-3000	12.0	20	-200	-40.0	35
DATA 8	2.4	0.4	-3000	12.0	20	-200	-40.0	35
DATA 9	2.4	0.4	-3000	12.0	20	-200	-40.0	35
DATA 10	2.4	0.4	-3000	12.0	20	-200	-40.0	35
DATA 11	2.4	0.4	-3000	12.0	20	-200	-40.0	35
DATA 12	2.4	0.4	-3000	12.0	20	-200	-40.0	35
DATA 13	2.4	0.4	-3000	12.0	20	-200	-40.0	35

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh min (uA)	Iol max (mA)	Iozh max (uA)	Iozl min (uA)	Ios max (mA)	Cout max (pF)
DATA 14	2.4	0.4	-3000	12.0	20	-200	-40.0	35
DATA 15	2.4	0.4	-3000	12.0	20	-200	-40.0	35
-HALTO	2.4	0.4	-224	3.54	NA	NA	-0.52	NA
-RESETO	2.4	0.4	-224	3.54	NA	NA	-0.52	NA
FC0	2.4	0.5	-375	2.95	45	-270	NA	40
FC1	2.4	0.5	-375	2.95	45	-270	NA	40
FC2	2.4	0.5	-375	2.95	45	-270	NA	40
-BEN	2.4	0.5	-1960	7.6	NA	NA	-30.0	NA
-TVPA	2.4	0.5	-2000	8.0	NA	NA	-30.0	NA
-BG	2.4	0.5	-624	2.75	NA	NA	-30.0	NA
-Y0	2.5	0.4	-400	4.0	NA	NA	-6.0	NA
-Y1	2.5	0.4	-400	4.0	NA	NA	-6.0	NA
-Y2	2.5	0.4	-400	4.0	NA	NA	-6.0	NA
-Y3	2.5	0.4	-400	4.0	NA	NA	-6.0	NA

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh min (uA)	Iol max (mA)	Iozh max (uA)	Iozl min (uA)	Ios max (mA)	Cout max (pF)
-Y4	2.5	0.4	-400	4.0	NA	NA	-6.0	NA
-Y5	2.5	0.4	-400	4.0	NA	NA	-6.0	NA
-Y6	2.5	0.4	-400	4.0	NA	NA	-6.0	NA
-Y7	2.5	0.4	-400	4.0	NA	NA	-6.0	NA
ADDR1	2.4	0.5	-400	3.2	20	-20	NA	25
ADDR2	2.4	0.5	-400	3.2	20	-20	NA	25
ADDR3	2.4	0.5	-400	3.2	20	-20	NA	25
ADDR4	2.4	0.5	-400	3.2	20	-20	NA	25
ADDR5	2.4	0.5	-400	3.2	20	-20	NA	25
ADDR6	2.4	0.5	-400	3.2	20	-20	NA	25
ADDR7	2.4	0.5	-400	3.2	20	-20	NA	25
ADDR8	2.4	0.5	-400	3.2	20	-20	NA	25
ADDR9	2.4	0.5	-400	3.2	20	-20	NA	25
ADDR10	2.4	0.5	-400	3.2	20	-20	NA	25

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh min (uA)	Iol max (mA)	Iozh max (uA)	Iozl min (uA)	Ios max (mA)	Cout max (pF)
ADDR11	2.4	0.5	-400	3.2	20	-20	NA	25
ADDR12	2.4	0.5	-400	3.2	20	-20	NA	25
ADDR13	2.4	0.5	-400	3.2	20	-20	NA	25
ADDR14	2.4	0.5	-400	3.2	20	-20	NA	25
ADDR15	2.4	0.5	-400	3.2	20	-20	NA	25
ADDR16	2.4	0.5	-400	3.2	20	-20	NA	25
ADDR17	2.4	0.5	-400	3.2	20	-20	NA	25
ADDR18	2.4	0.5	-400	3.2	20	-20	NA	25
ADDR19	2.4	0.5	-400	3.2	20	-20	NA	25
ADDR20	2.4	0.5	-400	3.2	20	-20	NA	25
ADDR21	2.4	0.5	-400	3.2	20	-20	NA	25
ADDR22	2.4	0.5	-400	3.2	20	-20	NA	25
ADDR23	2.4	0.5	-400	3.2	20	-20	NA	25
READ	2.4	0.5	-360	4.9	60	-420	NA	55

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh min (uA)	Iol max (mA)	Iozh max (uA)	Iozl min (uA)	Ios max (mA)	Cout max (pF)
-VMA	2.4	0.5	-400	5.3	20	-20	NA	25
-LDS	2.4	0.5	-400	5.3	20	-20	NA	25
-UDS	2.4	0.5	-400	5.3	20	-20	NA	25
E	2.4	0.5	-400	3.2	NA	NA	NA	NA
-AS	2.4	0.5	-599	4.6	-226	-770	NA	75
-IPL0	2.5	0.4	-594	11.14	-227	-957	-30.6	55
-IPL1	2.5	0.4	-594	11.14	-227	-957	-30.6	55
-IPL2	2.5	0.4	-594	11.14	-227	-957	-30.6	55
-EN	2.5	0.4	-400	4.0	NA	NA	-15.0	NA
CLKB	2.5	0.5	-1000	20.0	NA	NA	-40.0	NA
-CLKB	2.5	0.5	-1000	20.0	NA	NA	-40.0	NA
PCLK	2.5	0.5	-1000	20.0	NA	NA	-40.0	NA

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6.3.3 Dynamic parameters. The dynamic parameters appear with the dynamic test.

6.4 Module function. The module function is printed herein in Appendix A because of its length.

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6.4.1 Description of I/O pins.

Signal name	Pin number	I/O	Functional description
ADDR1	26	O	This 23 bit, unidirectional, three state output bus is capable of addressing eight megawords of data. During interrupt cycles, address lines ADDR1, ADDR2, and ADDR3 provide information about what level interrupt is being serviced while address lines ADDR4 through ADDR23 are all set to a high logic level.
ADDR2	77	O	
ADDR3	29	O	
ADDR4	28	O	
ADDR5	27	O	
ADDR6	78	O	
ADDR7	30	O	
ADDR8	32	O	
ADDR9	80	O	
ADDR10	82	O	
ADDR11	83	O	
ADDR12	84	O	
ADDR13	35	O	
ADDR14	36	O	
ADDR15	85	O	
ADDR16	34	O	
ADDR17	33	O	
ADDR18	81	O	
ADDR19	79	O	
ADDR20	76	O	
ADDR21	74	O	
ADDR22	86	O	
ADDR23	37	O	
DATA 0	94	I/O	This 16 bit, bidirectional, three state bus is the general purpose data path. During an interrupt acknowledge cycle, an external device supplies the vector number on data lines DATA 0 through DATA 7.
DATA 1	44	I/O	
DATA 2	42	I/O	
DATA 3	87	I/O	
DATA 4	38	I/O	
DATA 5	88	I/O	
DATA 6	39	I/O	
DATA 7	89	I/O	
DATA 8	41	I/O	
DATA 9	91	I/O	
DATA 10	92	I/O	
DATA 11	43	I/O	
DATA 12	93	I/O	
DATA 13	45	I/O	
DATA 14	95	I/O	
DATA 15	46	I/O	

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6.4.1 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Functional description
-AS	19	O	This active low output signal indicates that there is a valid address on the address bus. A 10 KOhm pull-up resistor is internally connected on the module to this signal to provide a valid output state while the 68000 is in the high impedance mode.
READ	68	O	This output signal defines the data bus transfer as a read (high) or write (low) cycle. The READ signal also works in conjunction with the upper and lower data strobes. This signal is three stateable.
-UDS -LDS	24 73	O O	These active low signals indicate the validity of the data on the data bus. This signal is three stateable.
-TVPA	72	O	This active low output pin is used solely as a test point for the 68000 -VPA input signal. It should never be used to drive the -VPA input of the 68000.
-BG	66	O	This active low output indicates to all other potential bus master devices that the NAH will release control of the bus at the end of the current bus cycle. It contains a 10 KOhm pull-up resistor to provide a valid initial state.

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6.4.1 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Functional description
-RESETO	9	O	This active low output is controlled both by the processor and by the -RESETI input. A processor generated reset resulting from the execution of a RESET instruction causes the -RESETO output to go low. This does not reset the processor; only the peripheral devices connected to this output are affected. The -RESETO output also goes low whenever the -RESETI input signal goes low. A 10 KOhm pull-up resistor is internally connected on the module to this signal.
FC 0 FC 1 FC 2	65 16 22	O O O	These function code outputs indicate the mode (user or supervisor) and the cycle type currently being executed. The information indicated by the function code outputs is valid whenever Address Strobe (-AS) is active.
DECEN 1 -DECEN 2 -DECEN 3	2 3 52	I I I	These signals are the three enable inputs for the three to eight line decoder (54LS138). For the decoder to be enabled, DECEN 1 must be high and both -DECEN 2 and -DECEN 3 must be low. A 10 KOhm pull-up resistor is internally connected on the module to DECEN 1.

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6.4.1 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Functional description
-VPA	13	I	This active low input indicates that the device or region addressed is a 68000 family device and that data transfer should therefore be synchronized with the enable (E) output signal. -VPA is derived by an external hardware decode of the address bus conditioned by Address Strobe (-AS). This input may also be used to indicate that the processor should use automatic vectoring for an interrupt. This signal is an input to the Programmable Array Logic (PAL) control decode function which generates the 68000 -VPA input signal. A 10 KOhm pull-up resistor is internally connected on the module to this signal.
DEC 0 DEC 1 DEC 2	54 4 53	I I I	These signals are the select inputs to the three to eight line decoder (54LS138) which can be used for 68000 function code (FC0-2) decoding if needed. The select inputs form a three bit binary code with DEC 2 being the most significant bit.
-Y0 -Y1 -Y2 -Y3 -Y4 -Y5 -Y6 -Y7	8 57 7 56 6 55 5 51	O O O O O O O O	These signals are the data outputs of the three to eight line decoder (54LS138). Each output is the active low decode of the binary value on the DEC 0-2 inputs. When the decoder is disabled, all outputs (-Y0-7) are high.

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6.4.1 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Functional description
E	23	O	This output signal is the standard enable signal common to all 68000 type peripheral devices. The period for this output is ten 68000 clock periods (six clocks low, four clocks high). Enable is generated by an internal ring counter which may come up in any state (for example, at power on, it is impossible to guarantee the phase relationship of E to the 68000 clock). Enable is a free running clock output and runs regardless of the state of the 68000.
-VMA	71	O	This active low output is used to indicate to 68000 peripheral devices that there is a valid address on the synchronized to enable (E). This signal is guaranteed only in response to a Valid Peripheral Address (-VPA) input which indicates that the addressed peripheral is a 68000 family device.

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6.4.1 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Functional description
-IPL0 -IPL1 -IPL2	69 70 21	I/O I/O I/O	These active low signals are the encoded state of the device(s) requesting an interrupt. These signals are inputs to a register (54LS377) that is used to synchronize them to the processor clock. When -ENINT is low, these signals are the outputs of the eight input priority encoder (54LS348) and no external logic can drive these lines. When -ENINT is high, the priority encoder outputs are in the high impedance state. The -IPL0-2 signals are then inputs which are driven by external logic. Level seven is the highest priority while level zero indicates that no interrupts are requested. The least significant bit is contained in -ILP0 and the most significant bit is contained in -IPL2. These inputs also have pull-up resistors (10 KOhm) connected on the module to provide a valid initial state, and to provide a no interrupt condition if the inputs are not used.

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6.4.1 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Functional description
-INT1 -INT2 -INT3 -INT4 -INT5 -INT6 -INT7	48 96 47 49 50 100 99	I I I I I I I	These active low inputs are used to supply interrupt signals to the 68000. They are encoded (54LS348) into a prioritized three bit 68000 compatible code (-IPL0-2) with seven having the highest priority. These inputs also have pullup resistors (10 KOhm) connected on the module to provide a valid initial state and to maintain a high level condition when they are not activated or left unused.
-BR	20	I	This active low input is wire-ORed with all other devices that could be bus masters. An active -BR input indicates to the processor that some other device desires to become the bus master. The alternate device pulls this input low to request the bus. A 10 KOhm pull-up resistor is internally connected on the module to this signal.
-ENINT	97	I	This active low input enables the eight input priority encode outputs as described above. A 10 KOhm pull-up resistor is connected to this line to provide a valid logic high input when the priority encoder is not used.
-EN	98	O	This active low output indicates that the eight input priority encoder is active and no external interrupt signals (-INT1-7) are active.

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6.4.1 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Functional description
-CLKB CLKB PCLK	60 61 11	O O O	These system clock outputs are TTL compatible and buffered (via the 54S86) for external system use. CLKB is the same frequency and approximate phase as the processor clock. -CLKB is the complement of the processor clock. The PCLK output may be programmed by the -INPHASE input to be equivalent to the processor clock (low) or its complement (high).
-BEN	18	O	When active low, this output indicates that: 1. The external reset input signal (-RESETI) is high. 2. The 68000 is not executing a -RESET instruction. 3. The -BGACK input signal is high. 4. The -HALT signal is high. 5. If the -HALT signal is active, the -AS output signal is active low. When the -BEN signal is low, the data transceivers are enabled. The data transceivers are in the high impedance mode when the -BEN signal is high.

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6.4.1 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Functional description
-VI	62	I	This active low module input is used to enable the automatic vector interrupt function. -VI is derived from an external logic request for automatic vectoring, usually for use with 68000 peripheral devices. During an interrupt acknowledge cycle while the processor is fetching the vector, if -VI is asserted, the processor will assert -VMA and complete a normal 68000 read cycle. The processor will then use an internally generated vector that is a function only of the interrupt being serviced. This process in known as automatic vectoring. The seven automatic vectors are vector numbers 25 through 31 (decimal). -VI is an input to the control decode function which generates the 68000 -VPA input signal. With the internal pullup resistor (10 KOhm) on the module, a high level condition is quaranteed for this -VI signal when it is not activated or left unused.
-DTACK	63	I	This active low input indicates that the data transfer is completed. When the processor recognizes -DTACK during a read cycle, data is latched and the bus cycle terminated. When -DTACK is recognized during a write cycle, the bus cycle is terminated. A 10 KOhm pull-up resistor is internally connected on the module signal.

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6.4.1 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Functional description
CLK	64	I	The clock input is a TTL compatible signal that is internally buffered for development of the internal clocks needed by the processor and the system clock outputs. The maximum clock frequency is 8.0 MHz and the minimum is 2.0 MHz.
-BGACK	67	I	This active low input indicates that some other device has become the bus master. A 10 KOhm pull-up resistor is internally connected on the module to this signal. This signal should not be asserted until the following four conditions are met: 1. A Bus Grant (-BG) has been received. 2. Address Strobe (-AS) is inactive which indicates that the processor is not using the bus. 3. Data Transfer Acknowledge (-DTACK) is inactive which indicates that neither memory nor peripheral are using the bus. 4. Bus Grant Acknowledge (-BGACK) is inactive which indicates that no other device is still claiming bus mastership.

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6.4.1 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Functional description
-INPHASE	59	I	This input selects either the non-inverted or inverted version of the processor clock to appear at the PCLK output. When -INPHASE is low, the PCLK output will be the same as the processor clock. With the -INPHASE high, the PCLK output will be the complement of the processor clock.
-BERR	17	I	This active low input informs the processor that there is a problem with the bus cycle currently being executed. A 10 KOhm pull-up resistor is internally connected on the module to this signal.
-HALTI	14	I	This active low input causes the processor to stop at the completion of the current bus cycle. When the processor has been halted using this input, all control signals are inactive and all three state output lines are put in their high impedance state by the control decode function. The data bus buffers are also put into the high impedance state by the control decode function. A 10 KOhm pull-up resistor is internally connected on the module to this signal.

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6.4.1 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Functional description
-HALTO	12	O	This active low output is controlled by the -HALTI input and the processor. When driven by the processor, this signal indicates to external devices that the processor has stopped executing instructions. The -HALTO output also goes low when the -HALTI input signal is driven low (for example, during a power on reset). A 10 KOhm pull-up resistor is internally connected on the module signal.
-RESETI	58	I	This active low input initiates a system initialization sequence. The -RESETI input must be held low for at least 100 mS during power up to perform a processor initialize sequence. This input is also distributed to peripheral devices via the -RESETO output. A 10 KOhm pull-up resistor is internally connected on the module to this signal.

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6.5 Description of tests. The following tables and associated figures describe the tests to be performed.

(a) Isolation test - table II.

(b) Continuity test - table III.

(c) Vcc stress test - table IV.

(d) Ibd test - table V.

(e) Vic test - table VI.

(f) Ios test - table VII.

(g) Iih test - table VIII.

(h) Iil test - table IX.

(i) Voh test - table X, figure 8.

(j) Vol test - table XI, figure 8.

(k) Iozh test - table XII.

(l) Iozl test - table XIII.

(m) Icc test - table XIV.

(n) Functional test - table XV, figures 8 and 9.

(o) Dynamic test - tables XVI and XVII, figures 10 and 11.

(p) PSFC test - table XVIII.

(q) PSTA test - table XIX, figure 12.

(r) Output crosstalk tests - tables XX and XXI, figures 13 and 14.

(s) Output crosstalk duration test - table XXII, figure 15.

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(t) Cin test - table XXIII.

(u) Cout test - table XXIV.

(v) Disable time test - table XXV, figure 16.

(w) Threshold test - table XXVI.

(x) Noise margin test - table XXVII, figure 17.

6.6 Functional test vectors. Functional test vectors are not printed herein because of their extreme length; and, even when presented as a hard copy, they require conversion to on-line format. When the entire sequence of vector groups is needed for driving devices as part of inspection, a test program output should be generated in a form compatible with test system architecture (magnetic tape format usually preferred). For reference purposes, the Naval Weapons Support Center (Code 607) maintains a limited quantity of dated, hard copy printouts of the functional test vectors and these may be obtained upon request.

Preparing activity:
NAVY - SH

(Project 5963-N743)

Agent: NW

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TABLE II. Isolation test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
A	Isolation, frame	All pins	Frame	10	INF	MOhms	1-7

NOTES:

1. Isolation test tolerance table.

Condition	Value	Accuracy	Test condition
Megohm bridge voltage	100 V	± 10 percent	A
Isolation measurement (see note 6)	NA NA	± 500 KOhm ± 50 KOhm	A A

2. The ohmmeter used shall be item R06.

3. In the setup conditions in table II, 'A' shall correspond to the positive terminal of the megohm bridge, and 'B' shall correspond to the negative terminal.

4. Warning: When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "DISCHARGE" before connecting or disconnecting the test leads.

5. When testing for frame isolation, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal of the megohm bridge.

6. The minimum end-of-life limit is 1 MOhm.

7. Abbreviations used are defined in 6.1.

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TABLE III. Continuity test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
B1	Continuity, Vcc	1	31	0.0	1.0	Ohm	1-4
B2	Continuity, GND	10	15				
B3	Continuity, GND	15	25				
B4	Continuity, GND	25	40				
B5	Continuity, GND	40	75				
B6	Continuity, GND	75	90	0.0	1.0	Ohm	1.4
B7	Continuity, Vcc	1	31	0.0	0.1	Ohm	1-4
B8	Continuity, GND	10	15				
B9	Continuity, GND	15	25				
B10	Continuity, GND	25	40				
B11	Continuity, GND	40	75				
B12	Continuity, GND	75	90	0.0	0.1	Ohm	1-4

NOTES.

1. Continuity test tolerance table.

Condition	Value	Accuracy	Test condition
Continuity measurement	NA NA	± 50 mOhm ± 5 mOhm	B1-B6 B7-B12

2. The ohmmeter used shall be item R07.

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NOTES: - Continued.

3. In the setup conditions in table III, 'A' shall correspond to the positive terminal of the ohmmeter, and 'B' shall correspond to the negative terminal.

4. Abbreviations used are defined in 6.1.

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TABLE IV. Power supply stress test.

Test cond	Power supply	Vin (V)	Connect Positive lead to	Connect Negative lead to	25 ± 5°C Initial limits	Notes
C	Vcc	7.0	1,31	10,15 25,40 75,90	Module shall pass all remaining tests.	1-5

NOTES:

1. Power supply stress test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	7.0 V	+0 mV,-70 mV	C
Power supply current limit	1.09 A	NA	C

2. There shall be no power supply voltage overshoot on this test.

3. Connect the power supplies to the module as indicated in table IV.

4. When performing the power supply stress test, no measurements are taken. This condition shall be maintained for a minimum of 5 seconds.

5. Abbreviations used are defined in 6.1.

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TABLE V. Input breakdown current test.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
D1	DATA 0	94	5.25	0.0	0.1	mA	1-3
D2	DATA 1	44					
D3	DATA 2	42					
D4	DATA 3	87					
D5	DATA 4	38					
D6	DATA 5	88					
D7	DATA 6	39					
D8	DATA 7	89					
D9	DATA 8	41					
D10	DATA 9	91					
D11	DATA 10	92					
D12	DATA 11	43					
D13	DATA 12	93					
D14	DATA 13	45					
D15	DATA 14	95					
D16	DATA 15	46					
D17	DECEN 1	2					
D18	-DECEN 2	3					
D19	-DECEN 3	52					
D20	DEC 0	54	5.25	0.0	0.1	mA	1-3

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TABLE V. Input breakdown current test - Continued.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D21	DEC 1	4	5.25	0.0	0.1	mA	1-3
D22	DEC 2	53					
D23	-BR	20					
D24	-BGACK	67					
D25	-BERR	17					
D26	-HALTI	14					
D27	-RESETI	58	5.25	0.0	0.1	mA	1-3
D28	-IPL0	69	5.25	0.0	0.12	mA	1-3
D29	-IPL1	70					
D30	-IPL2	21	5.25	0.0	0.12	mA	1-3
D31	-INT1	48	5.25	0.0	0.2	mA	1-3
D32	-INT2	96					
D33	-INT3	47					
D34	-INT4	49					
D35	-INT5	50					
D36	-INT6	100					
D37	-INT7	99					
D38	-ENINT	97	5.25	0.0	0.2	mA	1-3
D39	-VI	62	5.25	0.0	1.0	mA	1-3
D40	-VPA	13	5.25	0.0	1.0	mA	1-3

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TABLE V. Input breakdown current test - Continued.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D41	-DTACK	63	5.25	0.0	1.0	mA	1-3
D42	CLK	64	5.25	0.0	1.0	mA	1-3
D43	-INPHASE	59	5.25	0.0	1.0	mA	1-3

NOTES:

1. Input breakdown current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.25 V	± 52.5 mV	D1-D16
	5.25 V	+0.0 V, -52.5 mV	D17
	5.25 V	± 52.5 mV	D18-D22
	5.25 V	+0.0 V, -52.5 mV	D23-D41
	5.25 V	± 52.5 mV	D42
	5.25 V	+0.0 V, -52.5 mV	D43
Power supply current limit	1.09 A	NA	D1-D43
Forcing voltage, (Vin)	5.25 V	± 52.5 mV	D1-D16
	5.25 V	+0.0 mV, -52.5 mV	D17
	5.25 V	± 52.5 mV	D18-D22
	5.25 V	+0.0 mV, -52.5 mV	D23-D41
	5.25 V	± 52.5 mV	D42
	5.25 V	+0.0 mV, -52.5 mV	D43
Pattern voltage high, (VH)	3.0 V	± 100 mV	D1-D43
low, (VL)	0.0 V	± 100 mV	D1-D43

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Input current measurement	NA	± 5 uA	D1-D27
	NA	± 6 uA	D28-D30
	NA	± 10 uA	D31-D38
	NA	± 50 uA	D39-D43

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table V and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VI. Input clamp diode voltage test.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
E1	DATA 0	94	-18.0	-1.5	-0.4	V	1-3
E2	DATA 1	44					
E3	DATA 2	42					
E4	DATA 3	87					
E5	DATA 4	38					
E6	DATA 5	88					
E7	DATA 6	39					
E8	DATA 7	89					
E9	DATA 8	41					
E10	DATA 9	91					
E11	DATA 10	92					
E12	DATA 11	43					
E13	DATA 12	93					
E14	DATA 13	45					
E15	DATA 14	95					
E16	DATA 15	46					
E17	-INT1	48					
E18	-INT2	96					
E19	-INT3	47					
E20	-INT4	49	-18.0	-1.5	-0.4	V	1-3

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TABLE VI. Input clamp diode voltage test - Continued.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
E21	-INT5	50	-18.0	-1.5	-0.4	V	1-3
E22	-INT6	100					
E23	-INT7	99					
E24	DECEN 1	2					
E25	-DECEN 2	3					
E26	-DECEN 3	52					
E27	-IPL0	69					
E28	-IPL1	70					
E29	-IPL2	21					
E30	-BR	20					
E31	-BGACK	67					
E32	-BERR	17					
E33	DEC 0	54					
E34	DEC 1	4					
E35	DEC 2	53					
E36	-HALTI	14					
E37	-RESETI	58					
E38	-VI	62					
E39	-VPA	13					
E40	-DTACK	63	-18.0	-1.5	-0.4	V	1-3

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TABLE VI. Input clamp diode voltage test - Continued.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
E41	CLK	64	-18.0	-1.5	-0.4	V	1-3
E42	-ENINT	97	-18.0	-1.5	-0.4	V	1-3
E43	-INPHASE	59	-18.0	-1.2	-0.4	V	1-3

NOTES:

1. Input clamp diode voltage test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.75 V	± 47.5 mV	E1-E43
Power supply current limit	1.09 A	NA	E1-E43
Forcing current, (Iin)	-18 mA	± 1.8 mA	E1-E43
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 E1-E43 E1-E43
Input voltage measurement	 NA NA	 ± 55 mV ± 40 mV	 E1-E42 E43

2. The following procedure is required to perform this test:

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NOTES: - Continued.

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table VI and the voltage then measured.

3. Abbreviations used are defined in 6.1.

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TABLE VII. Output short circuit current test.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
F1	DATA 0	94	0.0	-225.0	-40.0	mA	1-4
F2	DATA 1	44					
F3	DATA 2	42					
F4	DATA 3	87					
F5	DATA 4	38					
F6	DATA 5	88					
F7	DATA 6	39					
F8	DATA 7	89					
F9	DATA 8	41					
F10	DATA 9	91					
F11	DATA 10	92					
F12	DATA 11	43					
F13	DATA 12	93					
F14	DATA 13	45					
F15	DATA 14	95					
F16	DATA 15	46	0.0	-225.0	-40.0	mA	1-4
F17	CLKB	61	0.0	-100.0	-40.0	mA	1-4
F18	-CLKB	60					
F19	PCLK	11	0.0	-100.0	-40.0	mA	1-4
F20	-Y0	8	0.0	-130.0	-6.0	mA	1-4

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TABLE VII. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F21	-Y1	57	0.0	-130.0	-6.0	mA	1-4
F22	-Y2	7					
F23	-Y3	56					
F24	-Y4	6					
F25	-Y5	55					
F26	-Y6	5					
F27	-Y7	51	0.0	-130.0	-6.0	mA	1-4
F28	-EN	98	0.0	-100.0	-15.0	mA	1-4
F29	-TVPA	72	0.0	-250.0	-30.0	mA	1-4
F30	-BEN	18	0.0	-130.4	-30.0	mA	1-4
F31	-HALTO	12	0.0	-552.6	-500.0	mA	1-4
F32	-RESETO	9	0.0	-552.6	-500.0	mA	1-4
F33	-IPL0	69	0.0	-130.95	-30.6	mA	1-4
F34	-IPL1	70					
F35	-IPL2	21	0.0	-130.95	-30.6	mA	1-4

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NOTES:

1. Output short circuit current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.25 V	± 52.5 mV	F1-F35
Power supply current limit	1.09 A	NA	F1-F35
Forcing voltage, (Vout)	0.0 V	± 20 mV	F1-F35
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 F1-F35 F1-F35
Output current measurement	NA NA NA NA NA NA NA NA	± 9.25 mA ± 3.0 mA ± 6.2 mA ± 4.25 mA ± 11.0 mA ± 5.02 mA ± 2.63 mA ± 5.02 mA	F1-F16 F17-F19 F20-F27 F28 F29 F30 F31,F32 F33-F35

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VII and the current then measured. The forcing voltage undershoot shall not exceed 100 mV.

3. When performing this test, only one output shall be tested at a time, and the duration of momentarily grounding each output shall not exceed 1 second.

4. Abbreviations used are defined in 6.1.

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TABLE VIII. Input current, high level, test.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G1	DATA 0	94	2.7	-20	20	uA	1-3
G2	DATA 1	44					
G3	DATA 2	42					
G4	DATA 3	87					
G5	DATA 4	38					
G6	DATA 5	88					
G7	DATA 6	39					
G8	DATA 7	89					
G9	DATA 8	41					
G10	DATA 9	91					
G11	DATA 10	92					
G12	DATA 11	43					
G13	DATA 12	93					
G14	DATA 13	45					
G15	DATA 14	95					
G16	DATA 15	46	2.7	-20	20	uA	1-3
G17	DEC 0	54	2.7	0	20	uA	1-3
G18	DEC 1	4					
G19	DEC 2	53					
G20	-DECEN 2	3	2.7	0	20	uA	1-3

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TABLE VIII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G21	-DECEN 3	52	2.7	0	20	uA	1-3
G22	-INT1	48	2.7	-268	-203	uA	1-3
G23	-INT2	96					
G24	-INT3	47					
G25	-INT4	49					
G26	-INT5	50					
G27	-INT6	100					
G28	-INT7	99	2.7	-268	-203	uA	1-3
G29	-IPL0	69	2.7	-288	-203	UA	1-3
G30	-IPL1	70					
G31	-IPL2	21	2.7	-288	-203	UA	1-3
G32	-ENINT	97	2.7	-268	-223	uA	1-3
G33	DECEN 1	2					
G34	-HALTI	14					
G35	-RESETI	58					
G36	-BR	20					
G37	-BGACK	67					
G38	-BERR	17	2.7	-268	-223	uA	1-3
G39	-INPHASE	59	2.7	-268	-193	uA	1-3
G40	-VI	62	2.4	-300	-246	uA	1-3

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TABLE VIII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G41	-VPA	13	2.4	-300	-246	uA	1-3
G42	-DTACK	63	2.4	-300	-246	uA	1-3
G43	CLK	64	2.4	0	25	uA	1-3

NOTES:

1. Input current, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.25 V	± 52.5 mV	G1-G43
Power supply current limit	1.09 A	NA	G1-G43
Forcing voltage, (Vih)	2.7 V 2.4 V	± 135 mV ± 120 mV	G1-G39 G40-G43
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	G1-G43 G1-G43

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Input current measurement	NA	± 2.0 uA	G1-G16
	NA	± 1.0 uA	G17-G21
	NA	± 3.25 uA	G22-G28
	NA	± 4.25 uA	G29-G31
	NA	± 2.25 uA	G32-G38
	NA	± 3.75 uA	G39
	NA	± 2.7 uA	G40-G42
	NA	± 1.25 uA	G43

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VIII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE IX. Input current, low level, test.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
H1	DATA 0	94	0.4	-200	0	uA	1-3
H2	DATA 1	44					
H3	DATA 2	42					
H4	DATA 3	87					
H5	DATA 4	38					
H6	DATA 5	88					
H7	DATA 6	39					
H8	DATA 7	89					
H9	DATA 8	41					
H10	DATA 9	91					
H11	DATA 10	92					
H12	DATA 11	43					
H13	DATA 12	93					
H14	DATA 13	45					
H15	DATA 14	95					
H16	DATA 15	46	0.4	-200	0	uA	1-3
H17	CLK	64	0.4	-250	0	uA	1-3
H18	DEC 0	54	0.4	-380	-1	uA	1-3
H19	DEC 1	4					
H20	DEC 2	53	0.4	-380	-1	uA	1-3

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TABLE IX. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
H21	-DECEN 2	3	0.4	-380	-1	uA	1-3
H22	-DECEN 3	52	0.4	-380	-1	uA	1-3
H23	-INT1	48	0.4	-1270	-722	uA	1-3
H24	-INT2	96					
H25	-INT3	47					
H26	-INT4	49					
H27	-INT5	50					
H28	-INT6	100					
H29	-INT7	99	0.4	-1270	-722	uA	1-3
H30	-HALTI	14	0.4	-891	-492	uA	1-3
H31	-RESETI	58	0.4	-891	-492	uA	1-3
H32	DECEN 1	2	0.4	-891	-463	uA	1-3
H33	-VI	62	0.4	-761	-462	uA	1-3
H34	-VPA	13					
H35	-DTACK	63	0.4	-761	-462	uA	1-3
H36	-IPL0	69	0.4	-931	-542	uA	1-3
H37	-IPL1	70					
H38	-IPL2	21	0.4	-931	-542	uA	1-3
H39	-BR	20	0.4	-911	-562	uA	1-3
H40	-BGACK	67	0.4	-911	-562	uA	1-3

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Check the source to verify that this is the current version before use.

TABLE IX. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
H41	-BERR	17	0.4	-911	-562	uA	1-3
H42	-ENINT	97	0.4	-871	-582	uA	1-3
H43	-INPHASE	59	0.5	-2500	-1452	uA	1-3

NOTES:

1. Input current, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.25 V	± 52.5 mV	H1-H43
Power supply current limit	1.09 A	NA	H1-H43
Forcing voltage, (Vil)	0.4 V 0.5 V	± 20 mV ± 25 mV	H1-H42 H43
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	H1-H43 H1-H43

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Input current measurement	NA	± 10.0 uA	H1-H16
	NA	± 12.5 uA	H17
	NA	± 19.0 uA	H18-H22
	NA	± 27.4 uA	H23-H29
	NA	± 20.0 uA	H30,H31
	NA	± 21.4 uA	H32
	NA	± 15.0 uA	H33-H35
	NA	± 19.4 uA	H36-H38
	NA	± 17.5 uA	H39-H41
	NA	± 14.4 uA	H42
	NA	± 52.4 uA	H43

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table IX and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE X. Output voltage, high level, test.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
J1	DATA 0	94	-3.0	2.4	4.75	V	1-4
J2	DATA 1	44					
J3	DATA 2	42					
J4	DATA 3	87					
J5	DATA 4	38					
J6	DATA 5	88					
J7	DATA 6	39					
J8	DATA 7	89					
J9	DATA 8	41					
J10	DATA 9	91					
J11	DATA 10	92					
J12	DATA 11	43					
J13	DATA 12	93					
J14	DATA 13	45					
J15	DATA 14	95					
J16	DATA 15	46	-3.0	2.4	4.75	V	1-4
J17	-TVPA	72	-2.0	2.4	4.75	V	1-4
J18	-BEN	18	-1.96	2.4	4.75	V	1-4
J19	-HALTO	12	-0.224	2.4	4.75	V	1-4
J20	-RESETO	9	-0.224	2.4	4.75	V	1-4

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TABLE X. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
J21	FC0	65	-0.375	2.4	4.75	V	1-4
J22	FC1	16	-0.375	2.4	4.75	V	1-4
J23	FC2	22	-0.375	2.4	4.75	V	1-4
J24	-BG	66	-0.624	2.4	4.75	V	1-4
J25	-AS	19	-0.599	2.4	4.75	V	1-4
J26	READ	68	-0.36	2.4	4.75	V	1-4
J27	ADDR1	26	-0.4	2.4	4.75	V	1-4
J28	ADDR2	77					
J29	ADDR3	29					
J30	ADDR4	28					
J31	ADDR5	27					
J32	ADDR6	78					
J33	ADDR7	30					
J34	ADDR8	32					
J35	ADDR9	80					
J36	ADDR10	82					
J37	ADDR11	83					
J38	ADDR12	84					
J39	ADDR13	35					
J40	ADDR14	36	-0.4	2.4	4.75	V	1-4

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TABLE X. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
J41	ADDR15	85	-0.4	2.4	4.75	V	1-4
J42	ADDR16	34					
J43	ADDR17	33					
J44	ADDR18	81					
J45	ADDR19	79					
J46	ADDR20	76					
J47	ADDR21	74					
J48	ADDR22	86					
J49	ADDR23	37					
J50	E	23					
J51	-VMA	71					
J52	-LDS	73					
J53	-UDS	24	-0.4	2.4	4.75	V	1-4
J54	-EN	98	-0.4	2.5	4.75	V	1-4
J55	-Y0	8					
J56	-Y1	57					
J57	-Y2	7					
J58	-Y3	56					
J59	-Y4	6					
J60	-Y5	55	-0.4	2.5	4.75	V	1-4

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TABLE X. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
J61	-Y6	5	-0.4	2.5	4.75	V	1-4
J62	-Y7	51	-0.4	2.5	4.75	V	1-4
J63	CLKB	61	-1.0	2.5	4.75	V	1-4
J64	-CLKB	60					
J65	PCLK	11	-1.0	2.5	4.75	V	1-4
J66	-IPL0	69	-0.594	2.5	4.75	V	1-4
J67	-IPL1	70					
J68	-IPL2	21	-0.594	2.5	4.75	V	1-4

NOTES:

1. Output voltage, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.75 V	± 47.5 mV	J1-J68
Power supply current limit	1.09 A	NA	J1-J68

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Forcing current, (Ioh)	-3.0 mA	± 30.0 uA	J1-J16
	-2.0 mA	± 20.0 uA	J17
	-1.96 mA	± 19.6 uA	J18
	-0.224 mA	± 2.24 uA	J19,J20
	-0.375 mA	± 3.75 uA	J21-J23
	-0.624 mA	± 6.24 uA	J24
	-0.599 mA	± 5.99 uA	J25
	-0.36 mA	± 3.6 uA	J26
	-0.4 mA	± 4.0 uA	J27-J62
	-1.0 mA	± 10.0 uA	J63-J65
	-0.594 mA	± 5.94 uA	J66-J68
Pattern voltage high, (VH)	2.0 V	± 20 mV	J1-J68
low, (VL)	0.7 V	± 7 mV	J1-J68
Output voltage measurement	NA	± 118 mV	J1-J53
	NA	± 113 mV	J54-J68

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table X and the voltage then measured.

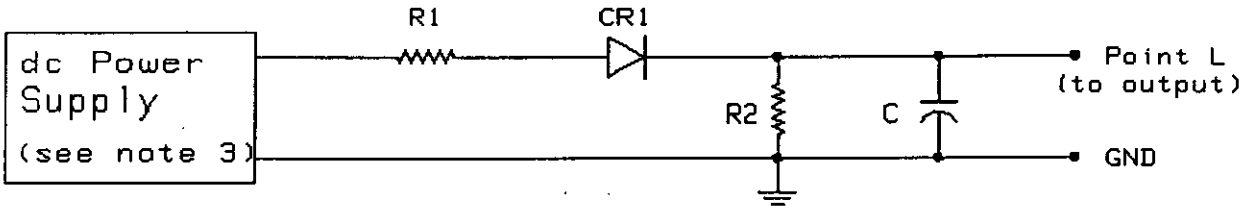
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NOTES: - Continued.

3. The load shall be as indicated on figure 8.

4. Abbreviations used are defined in 6.1.

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NOTES:

1. CR1 = 1N4148 or equivalent.

2. The load capacitance is defined as follows:

(a) C = 50 ± 5 pF (including probe and parasitic capacitance) for crosstalk and disable time tests.

(b) C = 150 ± 15 pF (including probe and parasitic capacitance) for sample dynamic tests.

(c) C = 300 ± 30 pF (including probe and parasitic capacitance) for qualification dynamic tests.

(d) The functional test shall be performed with only a 150 ± 15 pF load (including probe and parasitic capacitance) attached to each output.

(e) C not required for any other tests.

3. The load power supply voltage shall be 2.1 V ± 1 percent for the output disable time test and 2.5 V ± 1 percent for all other tests.

FIGURE 8. Logic load circuit.

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NOTES: - Continued.

4. Select R1 for a current flow of Iol when point L is held at Vol with the load power supply voltage set at 2.5 V ± 1 percent. The values of Iol for each output are specified below.

5. The value of R2 for each output load shall be as specified below.

Pin number	R2 1/ (ohms)	Iol (mA)	Vol (volts)
5,6,7,8,51,55,56, 57,98	6250	40.0	0.4
9,12	12000	3.54	0.4
11,60,61	2500	20.0	0.5
72	1206	8.0	0.5
16,22,65	6400	2.95	0.5
66	3865	2.75	0.5
18	1225	7.6	0.5
38,39,41,42,43,44, 45,46	800	12.0	0.4
87,88,89,91,92,93, 94,95,19	4027	4.6	0.5
21,69,70	4209	11.14	0.4
24,71,73	6000	5.3	0.5
23,26,27,28,29,30, 32,33	6000	3.2	0.5
34,35,36,37,74,76, 77,78,79,80,81,82, 83,84,85,86,68	6667	4.9	0.5

1/ All resistors are metal film, ± 1 percent, 1/4 W.

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TABLE XI. Output voltage, low level, test.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
K1	DATA 0	94	12.0	0.0	0.4	V	1-4
K2	DATA 1	44					
K3	DATA 2	42					
K4	DATA 3	87					
K5	DATA 4	38					
K6	DATA 5	88					
K7	DATA 6	39					
K8	DATA 7	89					
K9	DATA 8	41					
K10	DATA 9	91					
K11	DATA 10	92					
K12	DATA 11	43					
K13	DATA 12	93					
K14	DATA 13	45					
K15	DATA 14	95					
K16	DATA 15	46	12.0	0.0	0.4	V	1-4
K17	-HALTO	12	3.54	0.0	0.4	V	1-4
K18	-RESETO	9	3.54	0.0	0.4	V	1-4
K19	-Y0	8	4.0	0.0	0.4	V	1-4
K20	-Y1	57	4.0	0.0	0.4	V	1-4

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TABLE XI. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
K21	-Y2	7	4.0	0.0	0.4	V	1-4
K22	-Y3	56					
K23	-Y4	6					
K24	-Y5	55					
K25	-Y6	5					
K26	-Y7	51					
K27	-EN	98	4.0	0.0	0.4	V	1-4
K28	-IPL0	69	11.14	0.0	0.4	V	1-4
K29	-IPL1	70	11.14	0.0	0.4	V	1-4
K30	-IPL2	21	11.14	0.0	0.4	V	1-4
K31	ADDR1	26	3.2	0.0	0.5	V	1-4
K32	ADDR2	77					
K33	ADDR3	29					
K34	ADDR4	28					
K35	ADDR5	27					
K36	ADDR6	78					
K37	ADDR7	30					
K38	ADDR8	32					
K39	ADDR9	80					
K40	ADDR10	82	3.2	0.0	0.5	V	1-4

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TABLE XI. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
K41	ADDR11	83	3.2	0.0	0.5	V	1-4
K42	ADDR12	84					
K43	ADDR13	35					
K44	ADDR14	36					
K45	ADDR15	85					
K46	ADDR16	34					
K47	ADDR17	33					
K48	ADDR18	81					
K49	ADDR19	79					
K50	ADDR20	76					
K51	ADDR21	74					
K52	ADDR22	86					
K53	ADDR23	37					
K54	E	23	3.2	0.0	0.5	V	1-4
K55	-LDS	73	5.3	0.0	0.5	V	1-4
K56	-UDS	24					
K57	-VMA	71	5.3	0.0	0.5	V	1-4
K58	-AS	19	4.6	0.0	0.5	V	1-4
K59	READ	68	4.9	0.0	0.5	V	1-4
K60	-BEN	18	7.6	0.0	0.5	V	1-4

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TABLE XI. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
K61	FC0	65	2.95	0.0	0.5	V	1-4
K62	FC1	16					
K63	FC2	22	2.95	0.0	0.5	V	1-4
K64	-TVPA	72	8.0	0.0	0.5	V	1-4
K65	-BG	66	2.75	0.0	0.5	V	1-4
K66	CLKB	61	20.0	0.0	0.5	V	1-4
K67	-CLKB	60					
K68	PCLK	11	20.0	0.0	0.5	V	1-4

NOTES:

1. Output voltage, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.75 V	± 47.5 mV	K1-K68
Power supply current limit	1.09 A	NA	K1-K68

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:00Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Forcing current, (Iol)	12.0 mA	± 120.0 uA	K1-K16
	3.54 mA	± 35.4 uA	K17,K18
	4.0 mA	± 40.0 uA	K19-K27
	11.14 mA	± 111.4 uA	K28-K30
	3.2 mA	± 32.0 uA	K31-K54
	5.3 mA	± 53.0 uA	K55-K57
	4.63 mA	± 46.3 uA	K58
	4.9 mA	± 49.0 uA	K59
	7.6 mA	± 76.0 uA	K60
	2.95 mA	± 29.5 uA	K61-K63
	8.0 mA	± 80.0 uA	K64
	2.75 mA	± 27.5 uA	K65
	20.0 mA	± 200.0 uA	K66-K68
Pattern voltage high, (VH)	2.0 V	± 20 mV	K1-K68
low, (VL)	0.7 V	± 7 mV	K1-K68
Output voltage measurement	NA	± 20 mV	K1-K30
	NA	± 25 mV	K31-K68

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

Source: https://assist.dla.mil -- Downloaded: 20260-20T18:00Z
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NOTES: - Continued.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table XI and the voltage then measured.

3. The load shall be as indicated on figure 8.

4. Abbreviations used are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:00Z
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TABLE XII. Output leakage current, high level, test.

Test cond	Signal name	Pin no.	Forcing voltage (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
L1	DATA 0	94	2.7	-20	20	uA	1-3
L2	DATA 1	44					
L3	DATA 2	42					
L4	DATA 3	87					
L5	DATA 4	38					
L6	DATA 5	88					
L7	DATA 6	39					
L8	DATA 7	89					
L9	DATA 8	41					
L10	DATA 9	91					
L11	DATA 10	92					
L12	DATA 11	43					
L13	DATA 12	93					
L14	DATA 13	45					
L15	DATA 14	95					
L16	DATA 15	46	2.7	-20	20	uA	1-3
L17	ADDR1	26	2.4	-20	20	uA	1-3
L18	ADDR2	77					
L19	ADDR3	29					
L20	ADDR4	28	2.4	-20	20	uA	1-3

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:00Z
Check the source to verify that this is the current version before use.

TABLE XII. Output leakage current, high level, test - Continued.

Test cond	Signal name	Pin no.	Forcing voltage (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
L21	ADDR5	27	2.4	-20	20	uA	1-3
L22	ADDR6	78					
L23	ADDR7	30					
L24	ADDR8	32					
L25	ADDR9	80					
L26	ADDR10	82					
L27	ADDR11	83					
L28	ADDR12	84					
L29	ADDR13	35					
L30	ADDR14	36					
L31	ADDR15	85					
L32	ADDR16	34					
L33	ADDR17	33					
L34	ADDR18	81					
L35	ADDR19	79					
L36	ADDR20	76					
L37	ADDR21	74					
L38	ADDR22	86					
L39	ADDR23	37					
L40	-VMA	71	2.4	-20	20	uA	1-3

Source: https://assist.dla.mil -- Downloaded: 202610-20T18:00Z
Check the source to verify that this is the current version before use.

TABLE XII. Output leakage current, high level, test - Continued.

Test cond	Signal name	Pin no.	Forcing voltage (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
L41	-LDS	73	2.4	-20	20	uA	1-3
L42	-UDS	24	2.4	-20	20	uA	1-3
L43	READ	68	2.4	-20	60	uA	1-3
L44	FC0	65	2.4	-20	45	uA	1-3
L45	FC1	16					
L46	FC2	22	2.4	-20	45	uA	1-3
L47	-AS	19	2.4	-346	-226	uA	1-3

NOTES:

1. Output leakage current, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.25 V	± 52.5 mV	L1-L47
Power supply current limit	1.09 A	NA	L1-L47
Forcing voltage, (Voh)	2.7 V 2.4 V	± 135 mV ± 120 mV	L1-L16 L17-L47
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	L1-L47 L1-L47

Source: https://assist.dla.mil -- Downloaded: 202610-20T18:00Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Output current measurement	NA	± 2.0 uA	L1-L42
	NA	± 4.0 uA	L43
	NA	± 3.25 uA	L44-L46
	NA	± 6.0 uA	L47

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table XII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE XIII. Output leakage current, low level, test.

Test cond	Signal name	Pin no.	Vol (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
M1	DATA 0	94	0.4	-200	0	uA	1-3
M2	DATA 1	44					
M3	DATA 2	42					
M4	DATA 3	87					
M5	DATA 4	38					
M6	DATA 5	88					
M7	DATA 6	39					
M8	DATA 7	89					
M9	DATA 8	41					
M10	DATA 9	91					
M11	DATA 10	92					
M12	DATA 11	43					
M13	DATA 12	93					
M14	DATA 13	45					
M15	DATA 14	95					
M16	DATA 15	46	0.4	-200	0	uA	1-3
M17	ADDR1	26	0.4	-20	20	uA	1-3
M18	ADDR2	77					
M19	ADDR3	29					
M20	ADDR4	28	0.4	-20	20	uA	1-3

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TABLE XIII. Output leakage current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vol (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
M21	ADDR5	27	0.4	-20	20	uA	1-3
M22	ADDR6	78					
M23	ADDR7	30					
M24	ADDR8	32					
M25	ADDR9	80					
M26	ADDR10	82					
M27	ADDR11	83					
M28	ADDR12	84					
M29	ADDR13	35					
M30	ADDR14	36					
M31	ADDR15	85					
M32	ADDR16	34					
M33	ADDR17	33					
M34	ADDR18	81					
M35	ADDR19	79					
M36	ADDR20	76					
M37	ADDR21	74					
M38	ADDR22	86					
M39	ADDR23	37					
M40	-VMA	71	0.4	-20	20	uA	1-3

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TABLE XIII. Output leakage current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vol (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
M41	-LDS	73	0.4	-20	20	uA	1-3
M42	-UDS	24	0.4	-20	20	uA	1-3
M43	READ	68	0.4	-420	20	uA	1-3
M44	FC0	65	0.4	-270	20	uA	1-3
M45	FC1	16					
M46	FC2	22	0.4	-270	20	uA	1-3
M47	-AS	19	0.4	-770	-466	uA	1-3

NOTES:

1. Output leakage current, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.25 V	± 52.5 mV	M1-M47
Power supply current limit	1.09 A	NA	M1-M47
Forcing voltage, (Vol)	0.4 V	± 20 mV	M1-M47
Pattern voltage high, (VH)	3.0 V	± 100 mV	M1-M47
low, (VL)	0.0 V	± 100 mV	M1-M47

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Output current measurement	NA	± 10.0 uA	M1-M16
	NA	± 2.0 uA	M17-M42
	NA	± 22.0 uA	M43
	NA	± 14.5 uA	M44-M46
	NA	± 15.2 uA	M47

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions of table XIII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE XIV. Power supply current test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
N	Icc	1,31	36.33	726.64	mA	1-3

NOTES:

1. Power supply current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.25 V	± 52.5 mV	N
Power supply current limit	1.09 A	NA	N
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	N N
Power supply current measurement	NA	± 34.5 mA	N

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test and take a measurement at the pattern line with the maximum power supply current as indicated in table XIV.

3. Abbreviations used are defined in 6.1.

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TABLE XV. Functional test.

Test cond	Test description	Notes
P1	Functional test minimum power supply voltage maximum pattern rate	1-3
P2	Functional test maximum power supply voltage maximum pattern rate	1-3

NOTES:

1. Functional test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.75 V 5.25 V	± 47.5 mV ± 52.5 mV	P1 P2
Power supply current limit	1.09 A	NA	P1,P2
Pattern voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 120 mV ± 20 mV	 P1,P2 P1,P2
Output voltage verification high, (Voh) high, (Voh) low, (Vol)	 NA NA NA	 ± 117.5 mV ± 142.5 mV ± 25.0 mV	 P1 P2 P1,P2

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Use the appropriate pattern to verify the functional performance of the module as indicated in table XV using only a capacitive load of 100 pF (see figure 8). The output compare levels shall be as follows:

High level: $2.4\ V \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq 0.5\ V$

(c) The module inputs shall not violate the minimum input timing parameters referenced on figure 9.

3. Abbreviations used are defined in 6.1.

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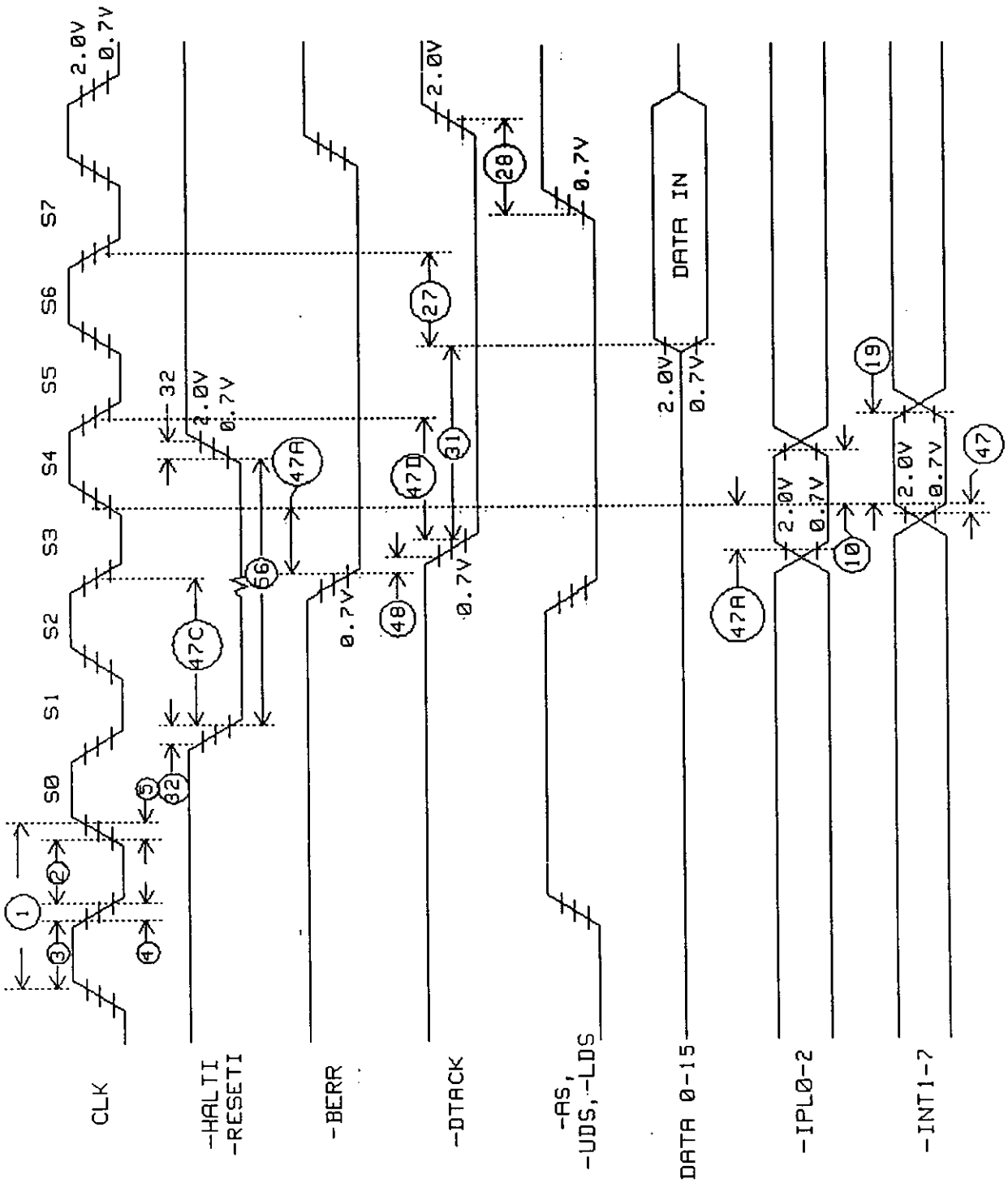


FIGURE 9. Input timing parameters for table XV.

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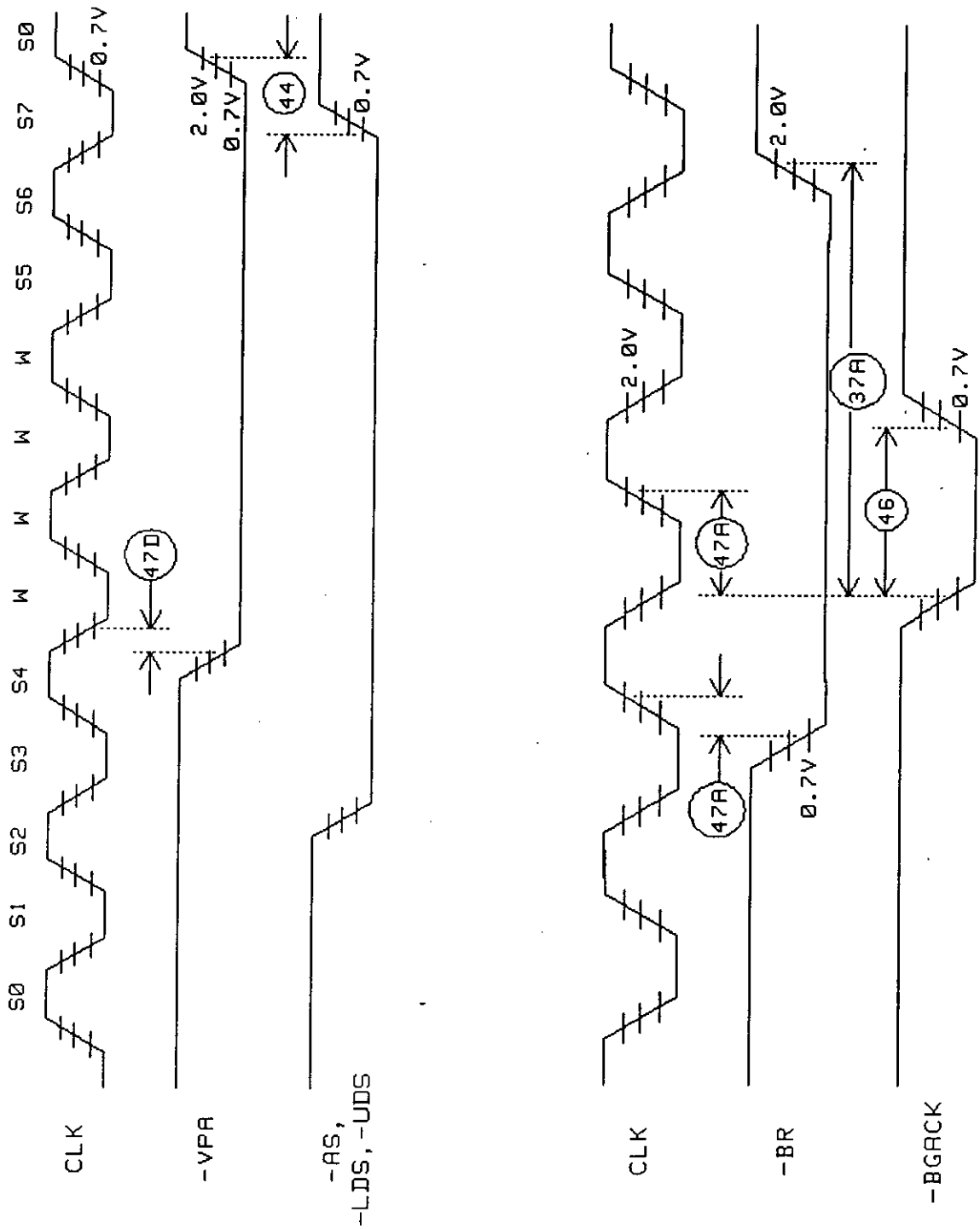


FIGURE 9. Input timing parameters for table XV - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:00Z
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NOTES:

1. (1) Clock period = 125 nS min, 500 nS max.

2. (2) Clock width low = 55 nS min.

3. (3) Clock width high = 55 nS min.

4. (4) Clock fall time = 10 nS max.

5. (5) Clock rise time = 10 nS max.

6. (10) -IPL0-2 to CLK high hold time = 66.5 nS min.

7. (19) -INT1-7 to CLK high hold time = 64.5 nS min.

8. (27) DATA 0-15 to CLK low setup time = 35 ns min. (See note 21.)

9. (28) -AS, -LDS, -UDS high to -DTACK high = 200 nS max.

10. (31) -DTACK low to DATA 0-15 (IN) setup time = 68 nS max. (See note 21.)

11. (32) -HALTI, -RESETI transition time = 200 nS max.

12. (37A) -BGACK low to -BR high (to prevent rearbitration) = 20 nS min.

13. (44) -AS, -LDS, -UDS high to -VPA high = 75 nS max.

14. (46) -BGACK pulse width low = 1.5 CLK periods min.

15. (47) -INT1-7 to CLK high setup time = 66 nS min. (See note 21.)

16. (47A) -IPL0-2, -BR, -BGACK, -BERR to CLK high setup time = 18 nS min. (See note 21.)

17. (47C) -HALTI, -RESETI to CLK low setup time = 53 nS min. (See note 21.)

18. (47D) -DTACK, -VPA low to CLK low setup time = 63 nS min. (See note 21.)

19. (48) -BERR low to -DTACK low = 20 nS min.

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NOTES: - Continued.

20. (56) -HALTI, -RESETI pulse width low = 10 CLK period min. For power up, the NAH module must be held in the reset state (-RESETI and -HALTI both low) for 100 mS min. for stabilization of the internal 68000 circuitry. After NAH module is powered up, #56 refers to the minimum pulse width required to reset the module. The clock input is required for the power up sequence, reference #1-5.

21. If the asynchronous setup time requirements are satisfied, the -DTACK low to DATA 0-15 setup time (#31) requirement can be ignored. The data must only satisfy the DATA 0-15 to CLK low setup time (#27) for the following bus cycle.

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TABLE XVI. Dynamic test.

Test cond	Test description	Notes
Q1	Dynamic test, sample maximum pattern rate minimum power supply	1-7
Q2	Dynamic test, sample maximum pattern rate maximum power supply	1-7
Q3	Dynamic test, qualification maximum pattern rate minimum power supply	1-7
Q4	Dynamic test, qualification maximum pattern rate maximum power supply	1-7

NOTES:

1. Dynamic test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.75 V 5.25 V	± 47.5 mV ± 52.5 mV	Q1,Q3 Q2,Q4
Power supply current limit	1.09 A	NA	Q1-Q4
Input voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 120 mV ± 20 mV	 Q1-Q4 Q1-Q4

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Output voltage verification			
high, (Voh)	NA	± 117.5 mV	Q1,Q3
high, (Voh)	NA	± 142.5 mV	Q2,Q4
low, (Vol)	NA	± 25.0 mV	Q1-Q4

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) The output compare levels shall be as follows:

High level: $Vref \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq Vref$

3. The load circuit for this test is specified on figure 8.

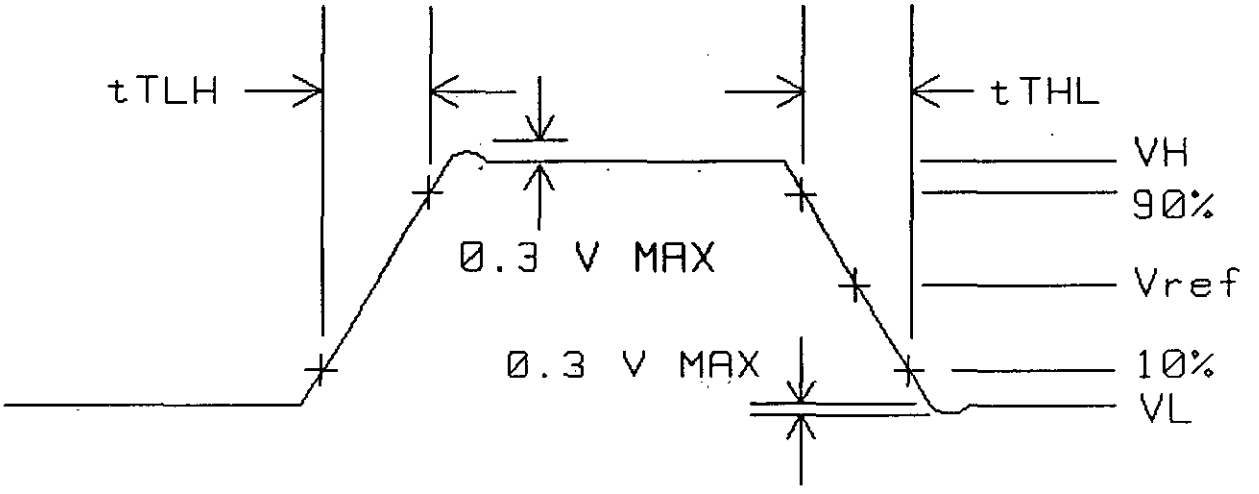
4. All of the input waveforms shall meet the requirements shown on figure 10.

5. This test shall be performed as indicated in table XVI. The dynamic parameters are found in table XVII.

6. Input/Output (I/O) waveforms are as referenced on figures 11a through 11h.

7. Abbreviations used are defined in 6.1.

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NOTES:

1. VH and VL are as specified in the test tolerance table.
2. Vref = 1.5 V.
3. Rise and fall times for the above waveform are as follows:

Input pin	tTLH	tTHL
2,3,4,13,14,17,20,21,38,39,41, 42,43,44,45,46,47,48,49,50,52, 53,54,58,62,63,64,67,69,70,87, 88,89,91,92,93,94,95,96,97,99, 100	15.0 ± 1.5 nS	6.0 ± 1.5 nS
59	2.5 ± 1.5 nS	2.5 ± 1.5 nS

FIGURE 10. Sample input waveform for table XVI.

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TABLE XVII. Dynamic parameters.

Ref	Parameter number	Parameter	0 to 60°C		25 ± 5°C Initial limits		Units	Fig
			Min	Max	Min	Max		
1	1	CLK frequency, F	2.0	8.0	2.0	8.0	MHz	11a
2	2	CLK pulse width low level, PWL	55.0	250.0	55.0	250.0	nS	11a
3	3	CLK pulse width high level, PWH	55.0	250.0	55.0	250.0	nS	11a
6	4-26	CLK low to ADDR1-23 valid, tPD	-	130.5	-	127.0	nS	11a,11b
6A	27-29	CLK high to FC0-2 valid, tPD	-	131.5	-	127.5	nS	11a,11b
9	30-32	CLK high to -AS, -LDS, -UDS low, tPHL	-	121.5	-	117.5	nS	11a,11b
11	33-36	ADDR1-23 to -AS, -LDS -UDS(read cycle) low/ -AS (write cycle), tPHL	30.0	-	30.0	-	nS	11a,11b
11A	37-40	FC0-2 valid to -AS, -LDS, -UDS (read cycle) low/-AS (write cycle), tPHL	60.0	-	60.0	-	nS	11a,11b
12	41-43	CLK low to -AS, -LDS, -UDS high, tPLH	-	130.5	-	127.0	nS	11a, 11b,11g
14	44-47	-AS, -LDS, -UDS pulse width low (read cycle) -AS (write cycle), PWL	240.0	-	240.0	-	nS	11a,11b

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TABLE XVII. Dynamic parameters - Continued.

Ref	Parameter number	Parameter	0 to 60°C		25 ± 5°C Initial limits		Units	Fig
			Min	Max	Min	Max		
14A	48,49	-LDS, -UDS pulse width low (write cycle), PWL	115.0	-	115.0	-	nS	11b
15	50-52	-AS, -LDS, -UDS, pulse width high, PWH	150.0	-	150.0	-	nS	11a,11b
17	53-55	-AS, -LDS, -UDS high to READ high, tPLH	40.0	-	40.0	-	nS	11a
18	57	CLK high to READ high, tPLH	-	131.5	-	127.5	nS	11a
20	58	CLK high to READ low, tPHL	-	131.5	-	127.5	nS	11b
20A	59	-AS low to READ low, tPHL	-	20.0	-	20.0	nS	11b
21	60	ADDR1-23 valid to READ low, tPHL	20.0	-	20.0	-	nS	11b
21A	61	FC0-2 valid to READ low, tPHL	60.0	-	60.0	-	nS	11b
22	62,63	READ low to -LDS, -UDS low (write cycle), tPHL	80.0	-	80.0	-	nS	11b
23	64-79	CLK low to DATA 0-15 out valid, tPD	-	160.5	-	152.0	nS	11b 11f,11g
25	80-95	-LDS, -UDS high to DATA 0-15 out invalid, tPD	32.0	-	32.0	-	nS	11b

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TABLE XVII. Dynamic parameters - Continued.

Ref	Parameter number	Parameter	0 to 60°C		25 ± 5°C Initial limits		Units	Fig
			Min	Max	Min	Max		
26	96-111	DATA 0-15 out valid to -LDS, -UDS, low (write cycle), tPHL	28.0	-	28.0	-	nS	11b
27	112-127	DATA 0-15 into CLK low setup time, tS	35.0	-	35.0	-	nS	11a, 11f, 11g
28	128-130	-AS, -LDS, -UDS high to -DTACK high, tH	-	181.5	-	185.0	nS	11a, 11b
29	131, 132	-LDS, -UDS high to DATA 0-15 in invalid (hold time), tH	0.0	-	0.0	-	nS	11a, 11f, 11g
30	133-135	-AS, -LDS, -UDS high to -BERR high, tH	0.0	-	0.0	-	nS	11a, 11b
31	136	-DTACK low to DATA 0-15 in (setup time), tS	-	68.0	-	73.0	nS	11a
33	141	CLK high to -BG low, tPHL	-	131.5	-	127.5	nS	11c
34	142	CLK high to -BG high, tPLH	-	131.5	-	127.5	nS	11c
35	143	-BR low to -BG low, tPHL	2.0	5.0	2.0	5.0	CLK Per.	11c
36	144	-BR high to -BG high, tPLH	2.0	5.0	2.0	5.0	nS	11c

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TABLE XVII. Dynamic parameters - Continued.

Ref	Parameter number	Parameter	0 to 60°C		25 ± 5°C Initial limits		Units	Fig
			Min	Max	Min	Max		
37	145	-BGACK low to -BG, tPLH	2.0	5.0	2.0	5.0	nS	11c
37A	146	-BGACK low to -BR high (skew time), tSKEW	-	1.0	-	1.0	CLK Per.	11c
39	147	-BG pulse width high, PWH	1.5	-	1.5	-	CLK Per.	11c
40	148	CLK low to -VMA low, tPHL	-	130.5	-	127.0	nS	11f,11g
41	149	CLK low to E transition	-	130.5	-	127.0	nS	11f,11g
43	152	-VMA low to E high, tPLH	200.0	-	200.0	-	nS	11f,11g
44	153-155	-AS, -LDS, -UDS high to -VPA high, tH	0.0	75.0	0.0	75.0	nS	11f,11g
46	183	-BGACK pulse width low, PWL	2.0	-	2.0	-	CLK Per.	11c
47A	184-188	-IPL0-2, -BR, -BERR to CLK high setup time, tS	18.0	-	18.0	-	nS	11a,11b
47A	189	-BGACK to CLK high setup time, tS	18.0	-	18.0	-	nS	11c

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TABLE XVII. Dynamic parameters - Continued.

Ref	Parameter number	Parameter	0 to 60°C		25 ± 5°C Initial limits		Units	Fig
			Min	Max	Min	Max		
47	190, 190A	-VPA, -VI to CLK low setup time, tS	63.0	-	63.0	-	nS	11f,11g
47A	191-197	-INT1-7 to CLK high setup time, tS	78.0	-	66.0	-	nS	11a,11b
49	199-201	-AS, -LDS, -UDS high from E low, tPLH	80.0	-	80.0	-	nS	11f,11g
50	202	E pulse width high, PWH	450.0	-	450.0	-	nS	11f,11g
51	203	E pulse width low, PWL	700.0	-	700.0	-	nS	11f,11g
53	205-220	DATA 0-15 hold time from CLK high, tH	.0.0	-	0.0	-	nS	11b
55	221-236	READ to DATA 0-15 out valid, tPD	32.0	-	32.0	-	nS	11b
56	237,238	-HALTI, -RESETI pulse width low, PWL	10.0	-	10.0	-	CLK Per.	11a,11b
47	239	-DTACK to CLK low setup time, tS	63.0	-	63.0	-	nS	11a,11b
47	240,241	-HALTI, -RESETI to CLK low setup time, tS	68.0	-	53.0	-	nS	11a,11b
10	242-244	-IPL0-2 to CLK high hold time, tH	66.5	-	62.5	-	nS	11a,11b
10	245-251	-INT1-7 to CLK high hold time, tH	64.5	-	60.5	-	nS	11a,11b

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TABLE XVII. Dynamic parameters - Continued.

Ref	Parameter number	Parameter	0 to 60°C		25 ± 5°C Initial limits		Units	Fig
			Min	Max	Min	Max		
60, 61	252,253	-HALTI, -RESETI high to -BEN low, tPHL	-	106.0	-	91.0	nS	11d
62	254	CLK high (-BGACK high) to -BEN low, tPHL	-	159.5	-	145.5	nS	11d
68	255	CLK high (to -AS high high) -HALTI (low) to -BEN low, tPLH	-	182.5	-	179.0	nS	11d
4	320	-VPA high to -TVPA high, tPLH	-	52.0	-	52.0	nS	11f
5	321	-VPA low to -TVPA low, tPHL	-	56.0	-	56.0	nS	11f
69	322,323	CLK high to CLKB, PCLK high, tPLH	-	67.5	-	63.5	nS	11e
71	324,325	CLK low to CLKB, PCLK low, tPHL	-	64.5	-	61.0	nS	11e
60 61	252A 253A	-HALTI (-AS high), -RESETI to -BEN low, tPLH	- -	97.0 97.0	- -	82.0 82.0	nS nS	11d 11d
62	254A	CLK high (-BGACK low) to -BEN low	-	155.5	-	141.5	nS	11d
74	326	-ENINT high to -EN high, tPLH	-	46.0	-	38.0	nS	11h
75	327	-ENINT low to -EN low, tPHL	-	63.0	-	51.0	nS	11h

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TABLE XVII. Dynamic parameters - Continued.

Ref	Parameter number	Parameter	0 to 60°C		25 ± 5°C Initial limits		Units	Fig
			Min	Max	Min	Max		
76	328-334	-INT1-7 high to -EN low, tPHL	-	69.0	-	56.0	nS	11h
77	335-341	-INT1-7 low to -EN high, tPLH	-	42.0	-	35.0	nS	11h
79	370-372	-ENINT to -IPL0-2 enable time, high output, tPZH	-	76.0	-	61.0	nS	11h
79	373-375	-ENINT to -IPL0-2 enable time, low output, tPZL	-	71.0	-	57.0	nS	11h
80	394-401	DEC 0-2 to -Y0-7, tPLH	-	60.0	-	44.0	nS	11h
82	402-409	DECEN 1, -DECEN 2, 3 to -Y0-7, tPLH	-	59.0	-	43.0	nS	11h
80	428-435	DEC 0-2 to -Y0-7, tPHL	-	92.0	-	65.0	nS	11h
81	436-443	-DECEN 2, 3 to -Y0-7, tPHL	-	76.0	-	54.0	nS	11h
70	444	CLK high to -CLKB low, tPHL	-	64.5	-	61.0	nS	11e
72	445	CLK low to -CLKB high, tPLH	-	67.5	-	63.5	nS	11e

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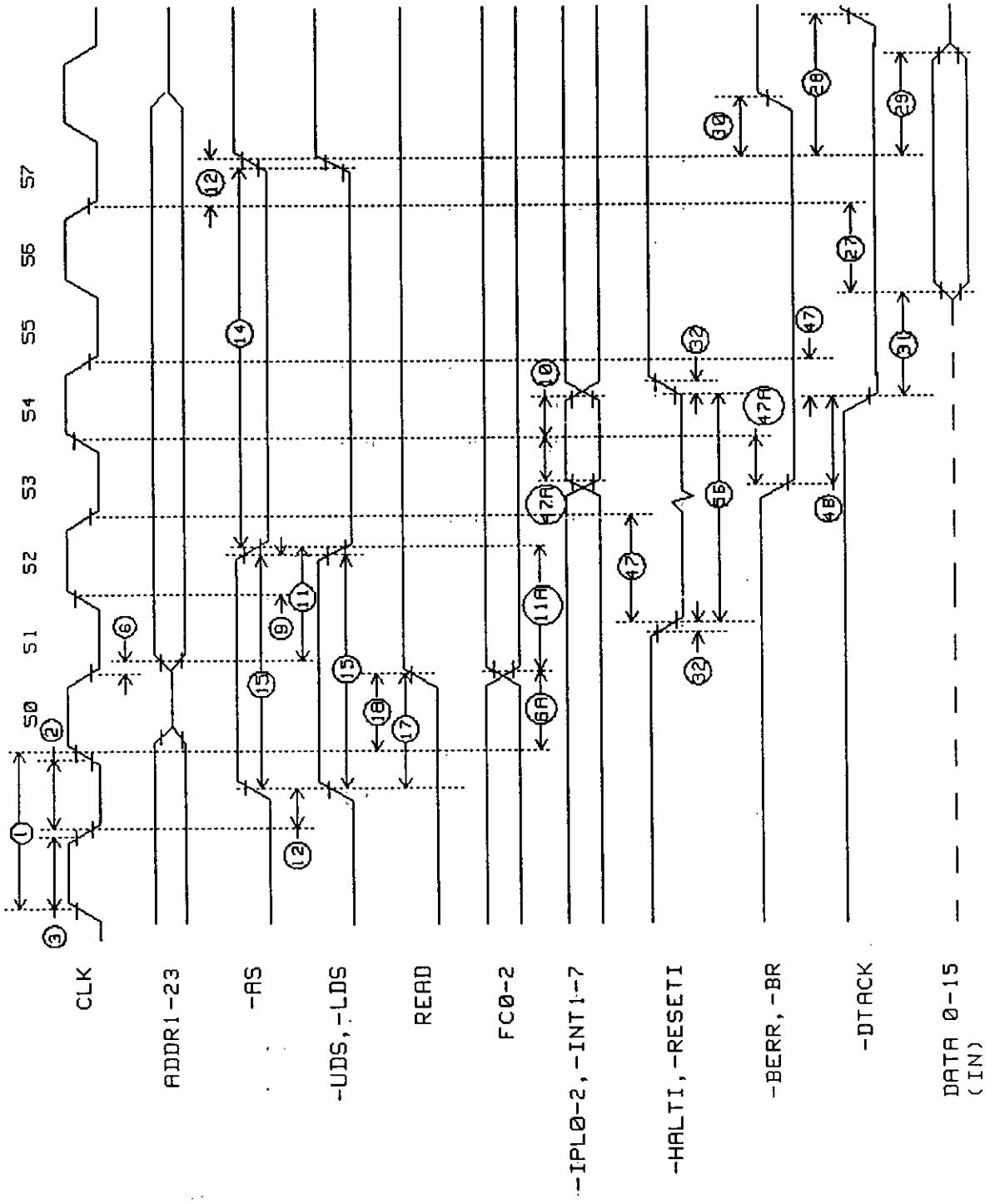


FIGURE 11a. Dynamic waveforms for table XVII.

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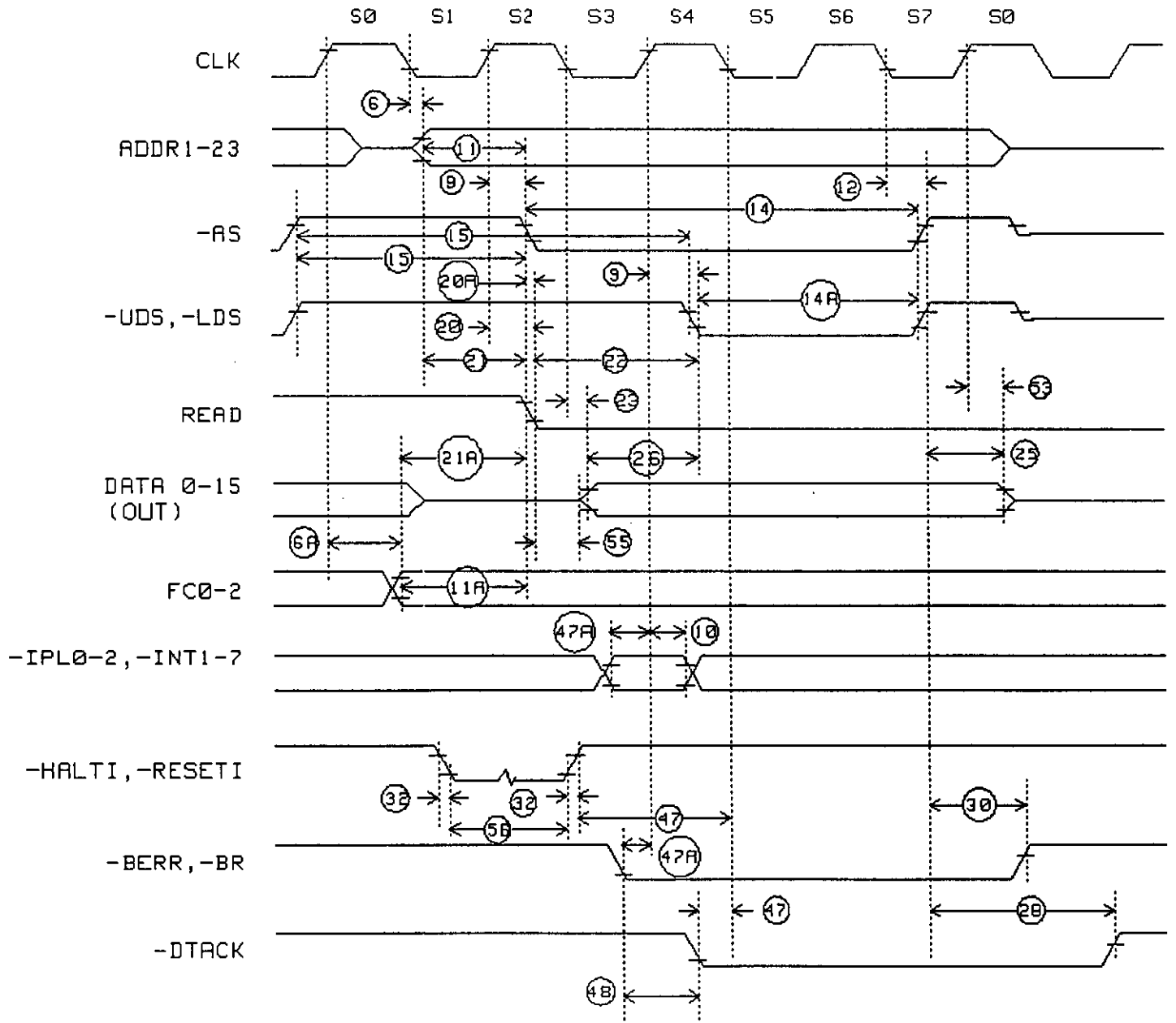


FIGURE 11b. Dynamic waveforms for table XVII.

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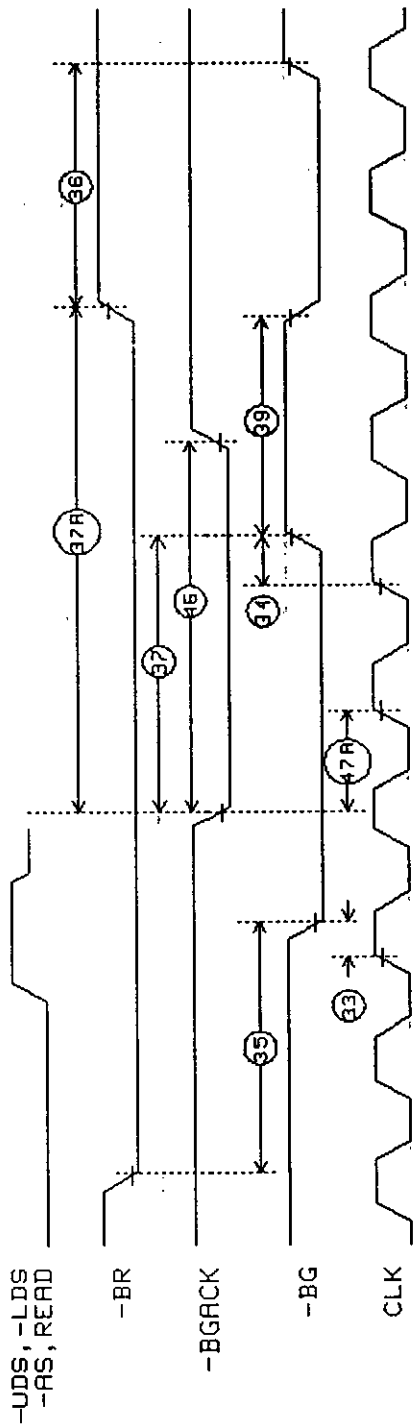


FIGURE 11c. Dynamic waveforms for table XVII.

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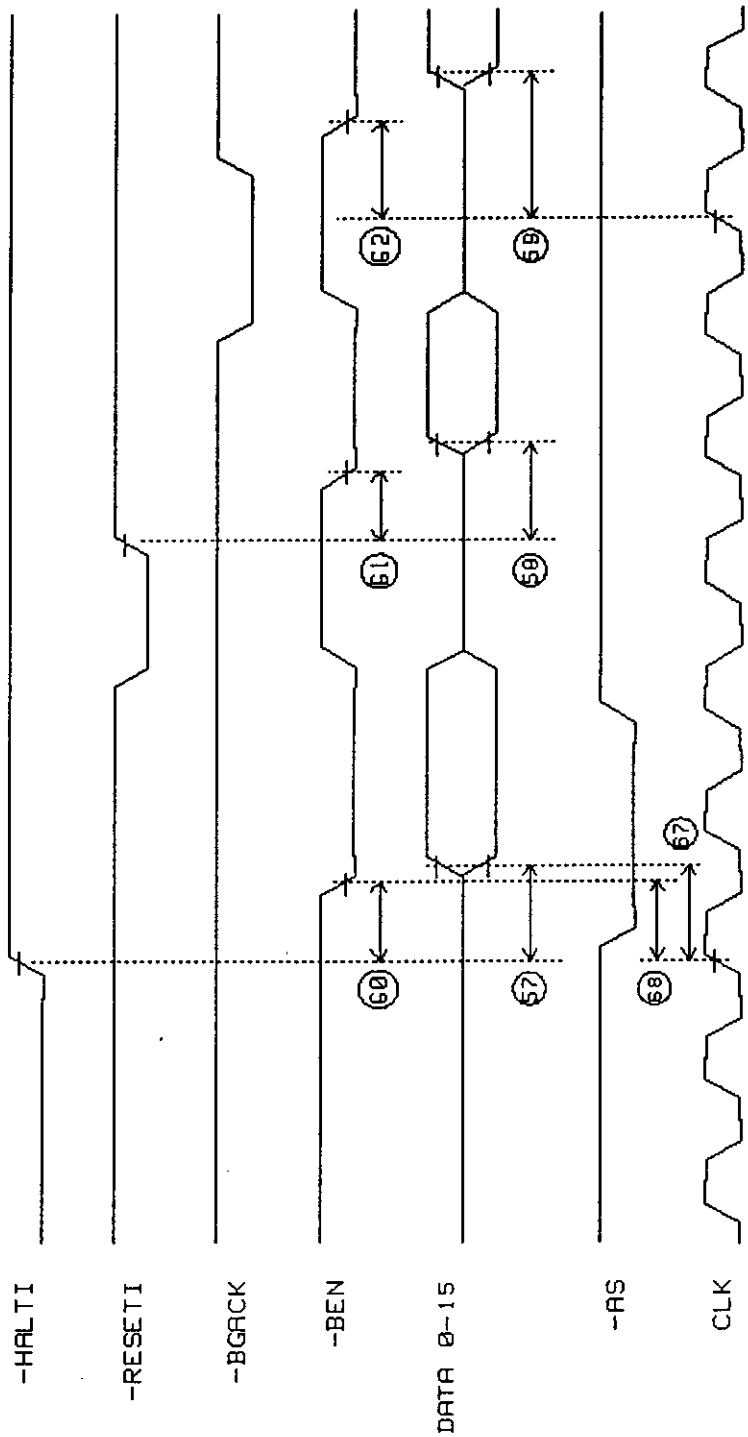


FIGURE 11d. Dynamic waveforms for table XVII.

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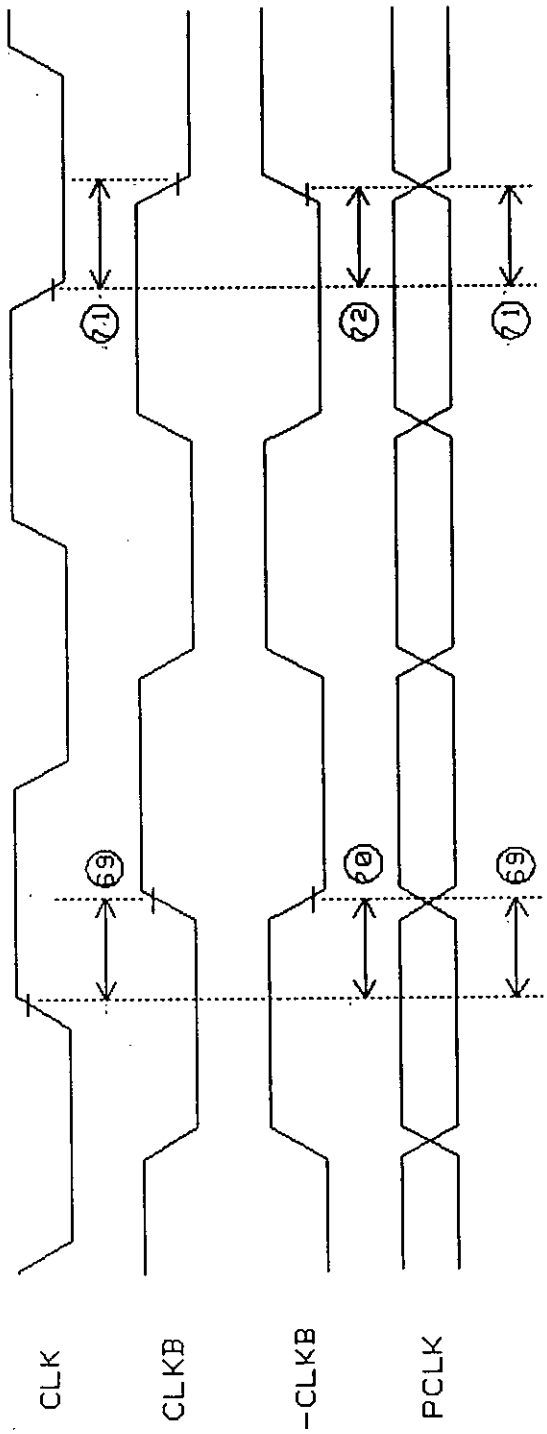


FIGURE 11e. Dynamic waveforms for table XVII.

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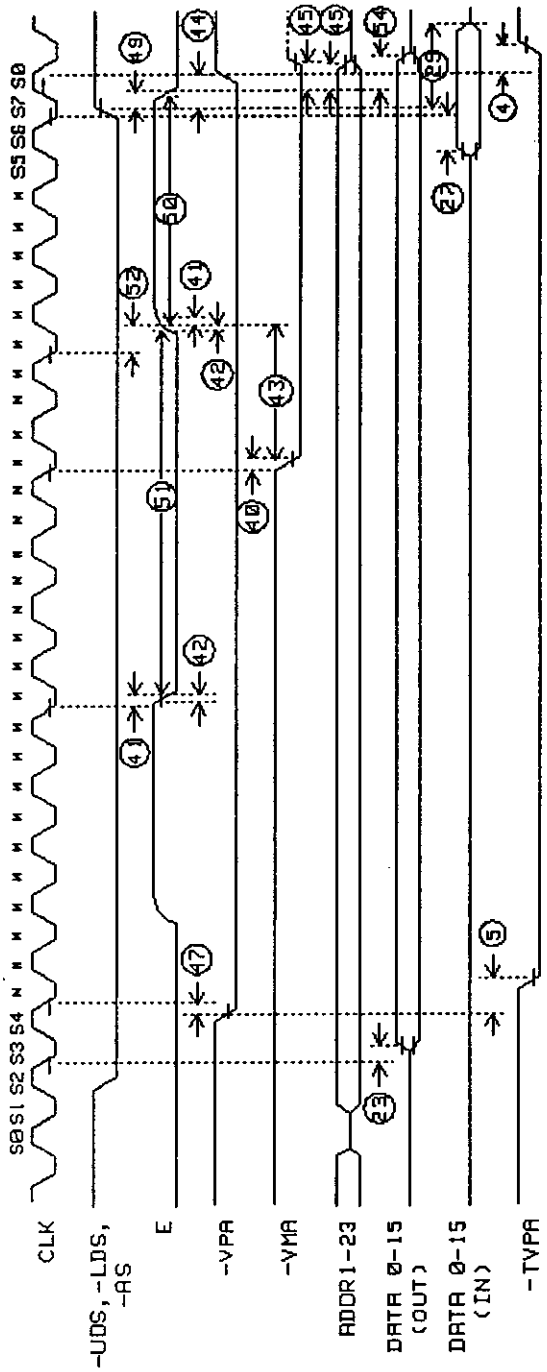


FIGURE 11f. Dynamic waveforms for table XVII.

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Check the source to verify that this is the current version before use.

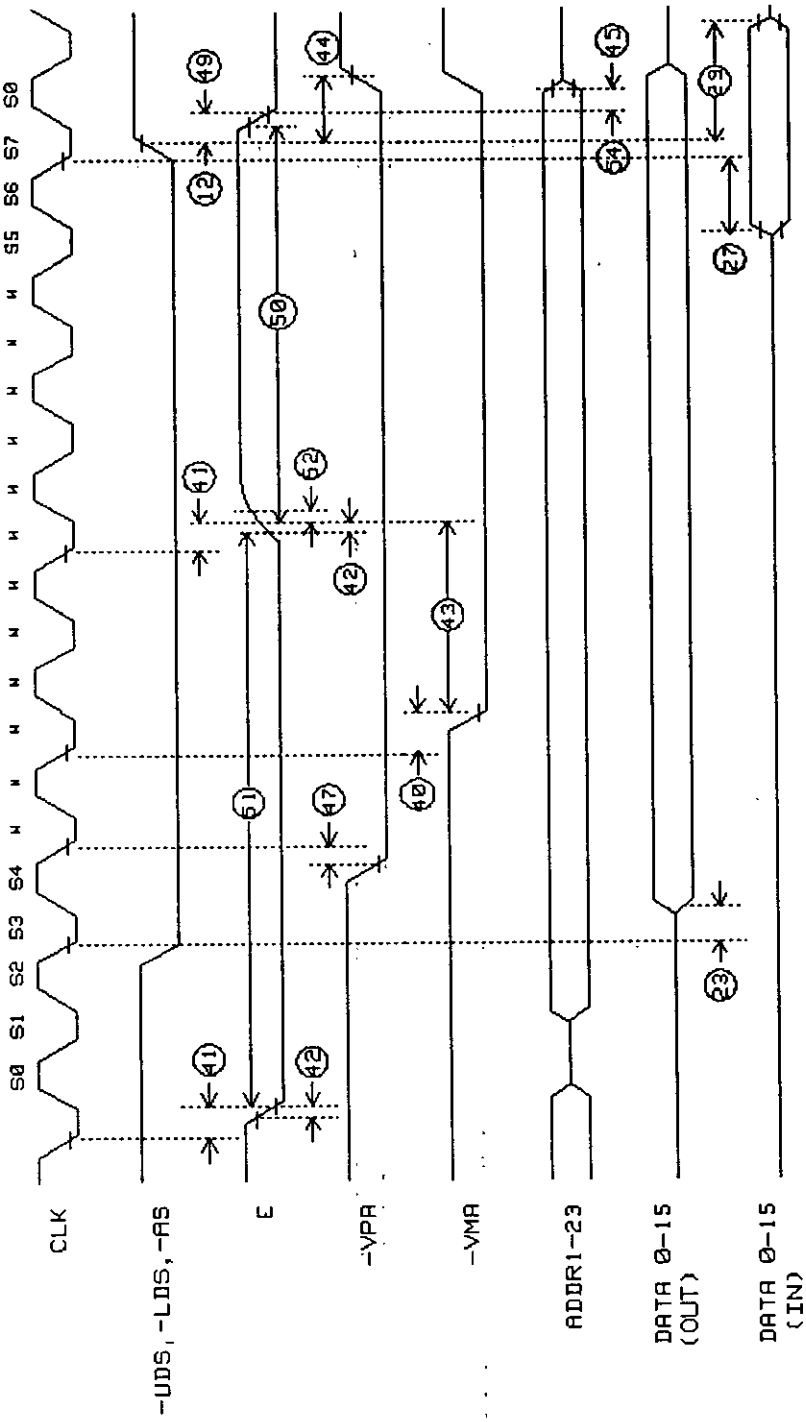


FIGURE 11g. Dynamic waveforms for table XVII.

Source: https://assist.dla.mil -- Downloaded: 20260-20T18:00Z
Check the source to verify that this is the current version before use.

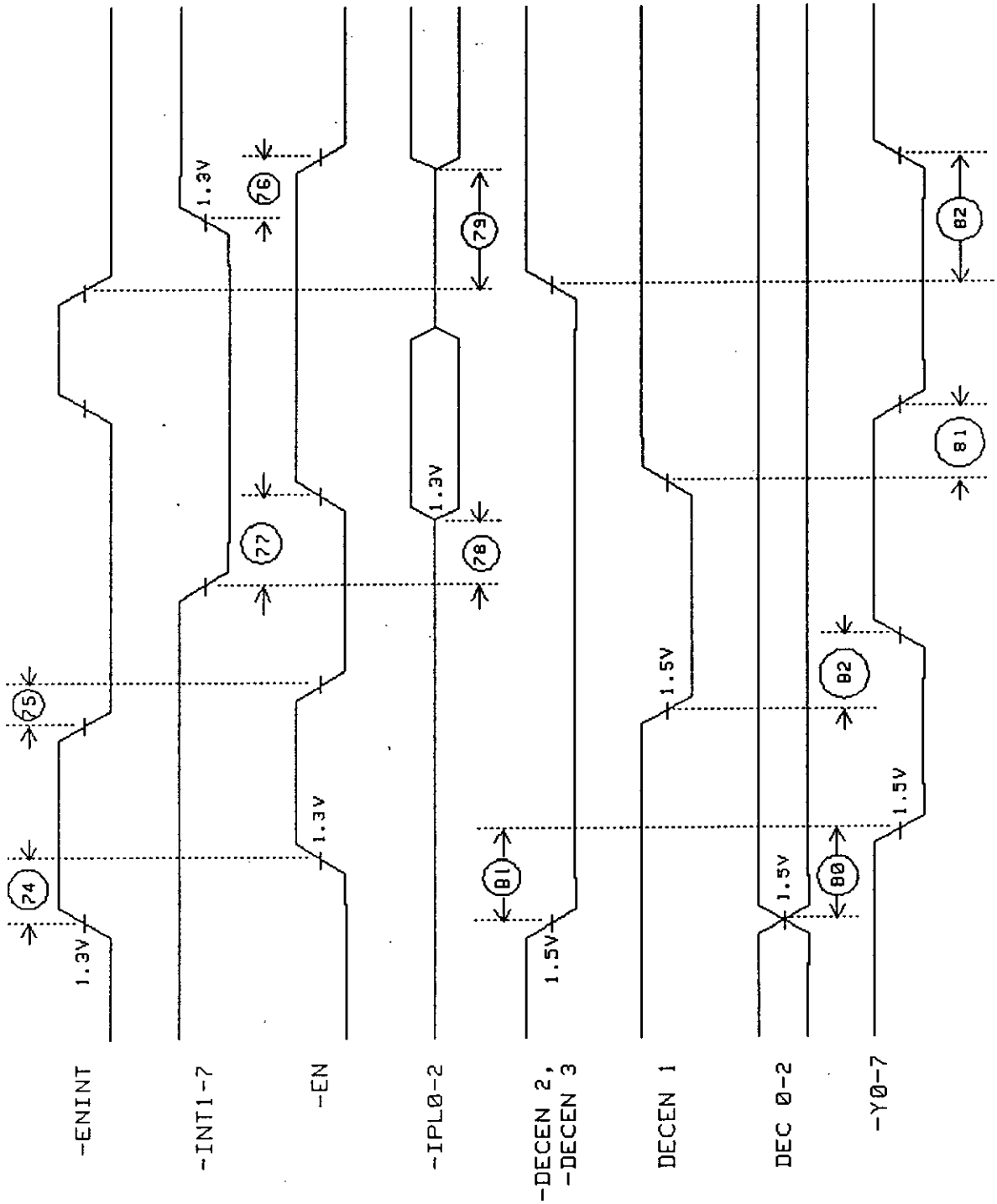


FIGURE 11h. Dynamic waveforms for table XVII.

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TABLE XVIII. Power supply filter capacitance test.

Test cond	Parameter	Pin numbers	25 ± 5°C Initial limits Min	Max	Units	Notes
R	PSFC	1,31	3.78	7.02	uF	1-3

NOTES:

1. Power supply filter capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance bridge frequency	1 KHz	± 100 Hz	R
Capacitance bridge voltage	0.3 Vp	± 30 mVp	R
Capacitance measurement	NA	± 162 nF	R

2. The capacitance bridge used in this test shall be item C01. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XVIII.

3. Abbreviations used are defined in 6.1.

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TABLE XIX. Power supply transient amplitude test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes	Fig
S	PSTA	1	0	400	mV	1-3	12

NOTES:

1. Power supply transient amplitude test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 250 mV	S
Power supply current limit	1.09 A	NA	S
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 S S
Power supply transient amplitude measurement	NA	± 20 mV	S

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table to an external source. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test and make the indicated measurement for the test condition shown in table XIX.

(c) All Vcc pins shall be shorted together and the scope point, probe 'A', shall be connected to the pin indicated in table XIX.

(d) Using the oscilloscope, D06, inspect for the most negative spike as shown on figure 12 at the pattern line with the maximum power supply current.

(e) The maximum probe capacitance is 15 pF. There shall be no other capacitance present on Vcc to ground.

3. Abbreviations used are defined in 6.1.

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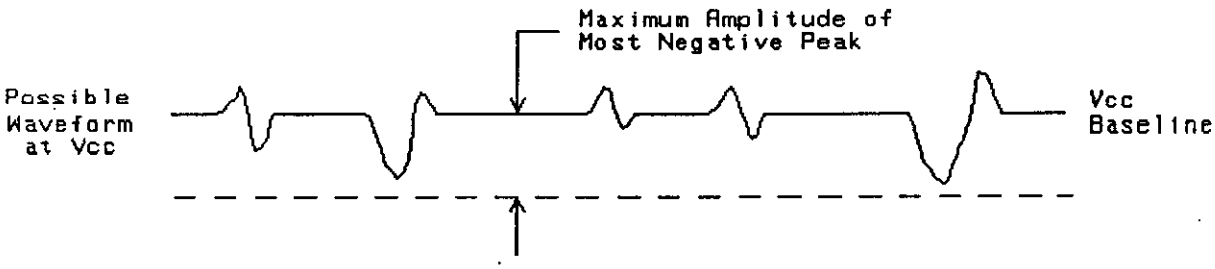


FIGURE 12. Power supply transient amplitude waveform for table XIX.

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TABLE XX. Output crosstalk, high level, test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes	Figs
T1	DATA 0	94	0	400	mV	1-5	13,14
T2	DATA 1	44					
T3	DATA 2	42					
T4	DATA 3	87					
T5	DATA 4	38					
T6	DATA 5	88					
T7	DATA 6	39					
T8	DATA 7	89					
T9	DATA 8	41					
T10	DATA 9	91					
T11	DATA 10	92					
T12	DATA 11	43					
T13	DATA 12	93					
T14	DATA 13	45					
T15	DATA 14	95					
T16	DATA 15	46					
T17	-HALTO	12					
T18	-RESETO	9					
T19	FC0	65					
T20	FC1	16	0	400	mV	1-5	13,14

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TABLE XX. Output crosstalk, high level, test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes	Figs
T21	FC2	22	0	400	mV	1-5	13,14
T22	-BEN	18					
T23	-TVPA	72					
T24	-BG	66					
T25	-Y0	8					
T26	-Y1	57					
T27	-Y2	7					
T28	-Y3	56					
T29	-Y4	6					
T30	-Y5	55					
T31	-Y6	5					
T32	-Y7	51					
T33	ADDR1	26					
T34	ADDR2	77					
T35	ADDR3	29					
T36	ADDR4	28					
T37	ADDR5	27					
T38	ADDR6	78					
T39	ADDR7	30					
T40	ADDR8	32	0	400	mV	1-5	13,14

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TABLE XX. Output crosstalk, high level, test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes	Figs
T41	ADDR9	80	0	400	mV	1-5	13,14
T42	ADDR10	82					
T43	ADDR11	83					
T44	ADDR12	84					
T45	ADDR13	35					
T46	ADDR14	36					
T47	ADDR15	85					
T48	ADDR16	34					
T49	ADDR17	33					
T50	ADDR18	81					
T51	ADDR19	79					
T52	ADDR20	76					
T53	ADDR21	74					
T54	ADDR22	86					
T55	ADDR23	37					
T56	READ	68					
T57	-VMA	71					
T58	-LDS	73					
T59	-UDS	24					
T60	E	23	0	400	mV	1-5	13,14

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TABLE XX. Output crosstalk, high level, test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes	Figs
T61	-AS	19	0	400	mV	1-5	13,14
T62	-IPL0	69					
T63	-IPL1	70					
T64	-IPL2	21	0	400	mV	1-5	13,14
T65	CLKB	61	0	500	mV	1-5	13,14
T66	-CLKB	60					
T67	PCLK	11	0	500	mV	1-5	13,14
T68	-EN	98	0	400	mV	1-5	13,14

NOTES:

1. Output crosstalk, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 0.25 V	T1-T68
Power supply current limit	1.09 A	NA	T1-T68
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 T1-T68 T1-T68

Source: https://assist.dla.mil -- Downloaded: 20261O-20T18:00Z
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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Output crosstalk measurement	NA NA NA	± 20 mV ± 25 mV ± 20 mV	T1-T64 T65-T67 T68

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) Use probe 'A' to verify the pattern voltage and pattern repetition rate of 1000 nS ± 200 nS, 50 percent duty cycle.

(d) For each output under test in table XX, connect probe 'B' and visually inspect the output for the most negative spike as indicated on figure 13.

(e) The maximum probe capacitance is 10 pF.

3. All outputs shall be loaded as specified on figure 8.

4. The test circuit shall be as indicated on figure 14.

5. Abbreviations used are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:00Z
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TABLE XXI. Output crosstalk, low level, test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes	Figs
U1	DATA 0	94	0	400	mV	1-5	13,14
U2	DATA 1	44					
U3	DATA 2	42					
U4	DATA 3	87					
U5	DATA 4	38					
U6	DATA 5	88					
U7	DATA 6	39					
U8	DATA 7	89					
U9	DATA 8	41					
U10	DATA 9	91					
U11	DATA 10	92					
U12	DATA 11	43					
U13	DATA 12	93					
U14	DATA 13	45					
U15	DATA 14	95					
U16	DATA 15	46					
U17	-HALTO	12					
U18	-RESETO	9					
U19	FC0	65					
U20	FC1	16	0	400	mV	1-5	13,14

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TABLE XXI. Output crosstalk, low level, test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes	Figs
U21	FC2	22	0	400	mV	1-5	13,14
U22	-BEN	18					
U23	-TVPA	72					
U24	-BG	66					
U25	-Y0	8					
U26	-Y1	57					
U27	-Y2	7					
U28	-Y3	56					
U29	-Y4	6					
U30	-Y5	55					
U31	-Y6	5					
U32	-Y7	51					
U33	ADDR1	26					
U34	ADDR2	77					
U35	ADDR3	29					
U36	ADDR4	28					
U37	ADDR5	27					
U38	ADDR6	78					
U39	ADDR7	30					
U40	ADDR8	32	0	400	mV	1-5	13,14

Source: https://assist.dla.mil -- Downloaded: 202610-20T18:00Z
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TABLE XXI. Output crosstalk, low level, test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes	Figs
U41	ADDR9	80	0	400	mV	1-5	13,14
U42	ADDR10	82					
U43	ADDR11	83					
U44	ADDR12	84					
U45	ADDR13	35					
U46	ADDR14	36					
U47	ADDR15	85					
U48	ADDR16	34					
U49	ADDR17	33					
U50	ADDR18	81					
U51	ADDR19	79					
U52	ADDR20	76					
U53	ADDR21	74					
U54	ADDR22	86					
U55	ADDR23	37					
U56	READ	68					
U57	-VMA	71					
U58	-LDS	73					
U59	-UDS	24					
U60	E	23	0	400	mV	1-5	13,14

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TABLE XXI. Output crosstalk, low level, test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes	Figs
U61	-AS	19	0	400	mV	1-5	13,14
U62	-IPL0	69					
U63	-IPL1	70					
U64	-IPL2	21	0	400	mV	1-5	13,14
U65	CLKB	61	0	500	mV	1-5	13,14
U66	-CLKB	60					
U67	PCLK	11	0	500	mV	1-5	13,14
U68	-EN	98	0	700	mV	1-5	13,14

NOTES:

1. Output crosstalk, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 0.25 V	U1-U68
Power supply current limit	1.09 A	NA	U1-U68
Pattern voltage			
high, (VH)	3.0 V	± 100 mV	U1-U68
low, (VL)	0.0 V	± 100 mV	U1-U68

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Output crosstalk measurement	NA NA NA	± 20 mV ± 25 mV ± 35 mV	U1-U64 U65-U67 U68

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) Use probe 'A' to verify the pattern voltage and pattern repetition rate of 1000 nS ± 200 nS, 50 percent duty cycle.

(d) For each output under test in table XXI, connect probe 'B' and visually inspect the output for the most positive spike as indicated on figure 13.

(e) The maximum probe capacitance is 10 pF.

3. All outputs shall be loaded as specified on figure 8.

4. The test circuit shall be as indicated on figure 14.

5. Abbreviations used are defined in 6.1.

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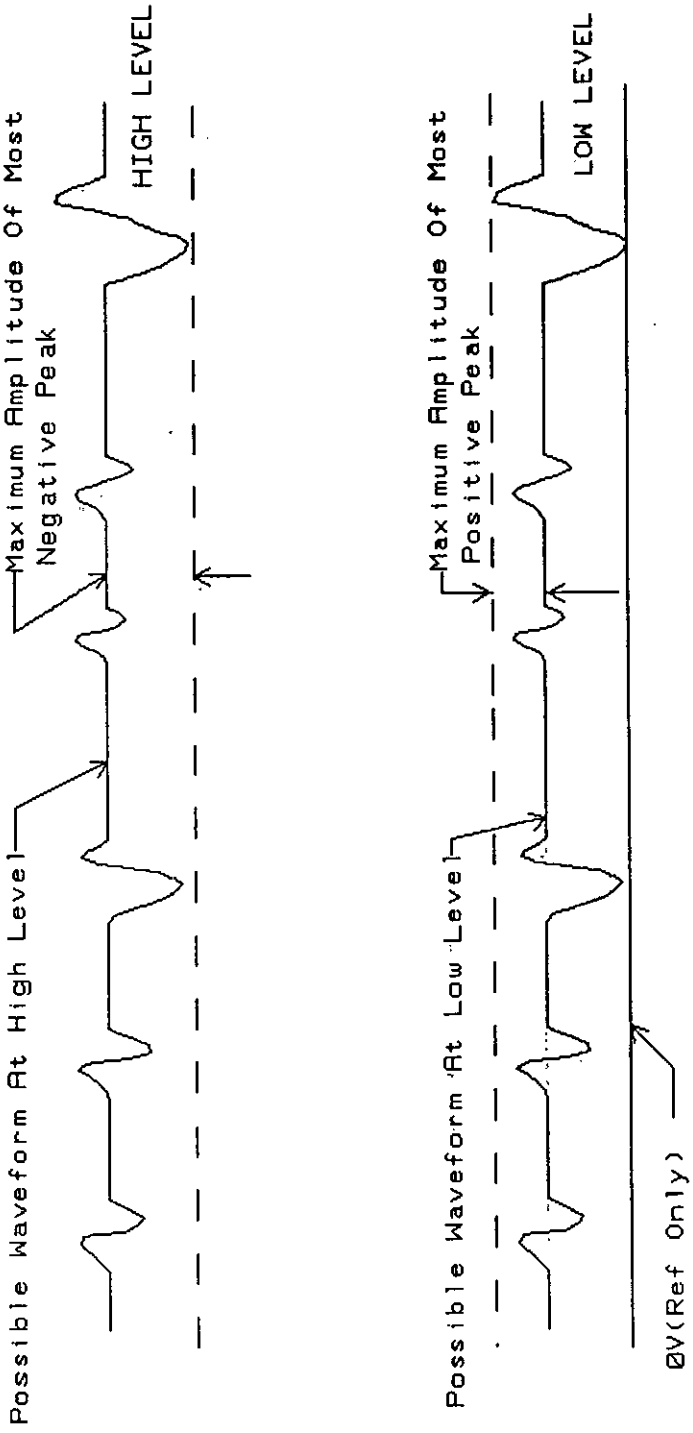


FIGURE 13. Output crosstalk, high and low level, test waveforms for tables XX and XXI.

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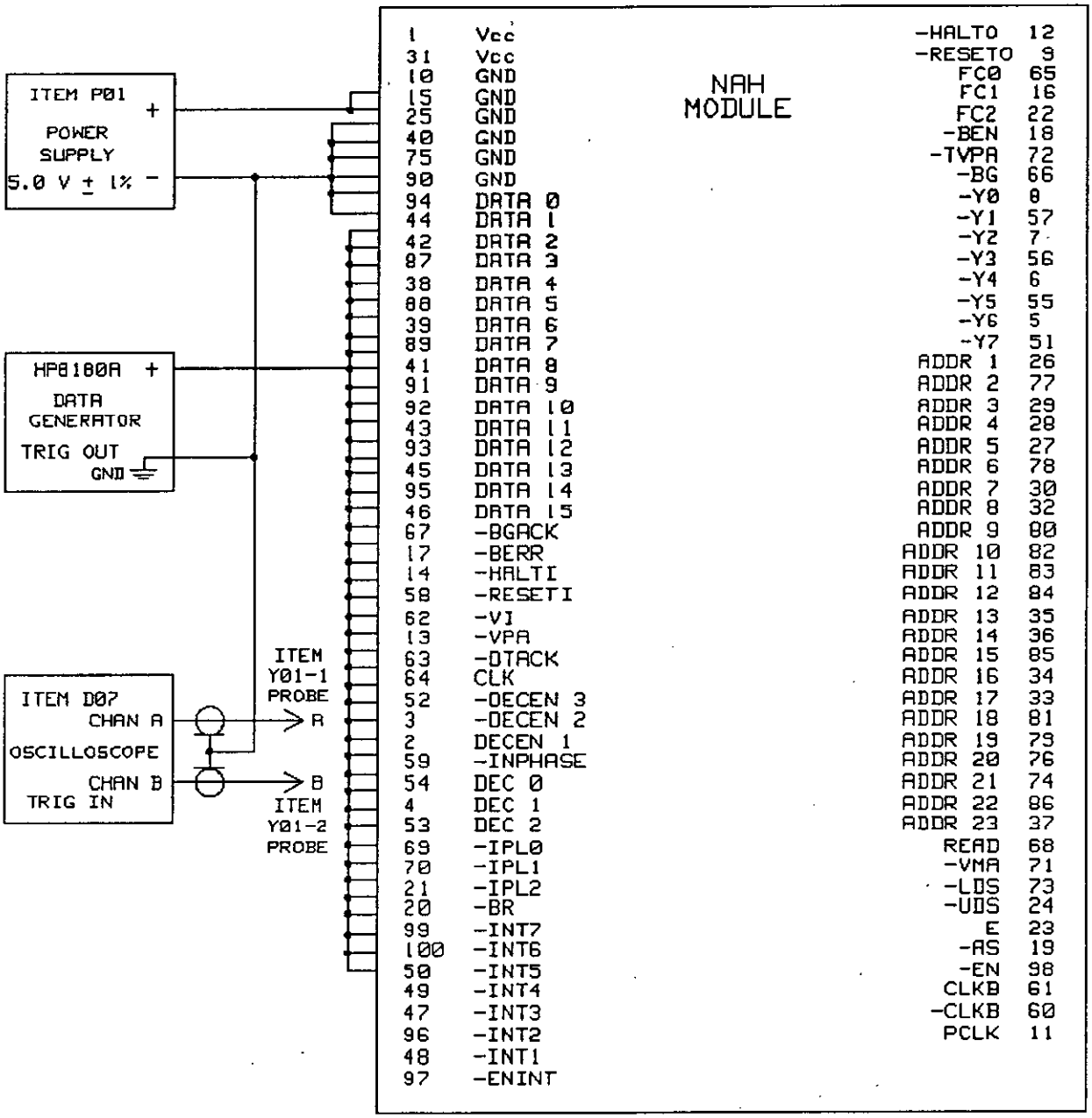


FIGURE 14. Output crosstalk, high and low level, test circuit for tables XX and XXI.

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TABLE XXII. Output crosstalk, duration, test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes	Figs
V	-EN	98	0	50	nS	1-4	15

NOTES:

1. Output crosstalk, duration, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 0.25 V	V
Power supply current limit	1.09 A	NA	V
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	V V
Output crosstalk measurement	NA	± 2.5 nS	V

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) The duration is measured from the triggering edge of the input as shown on figure 15.

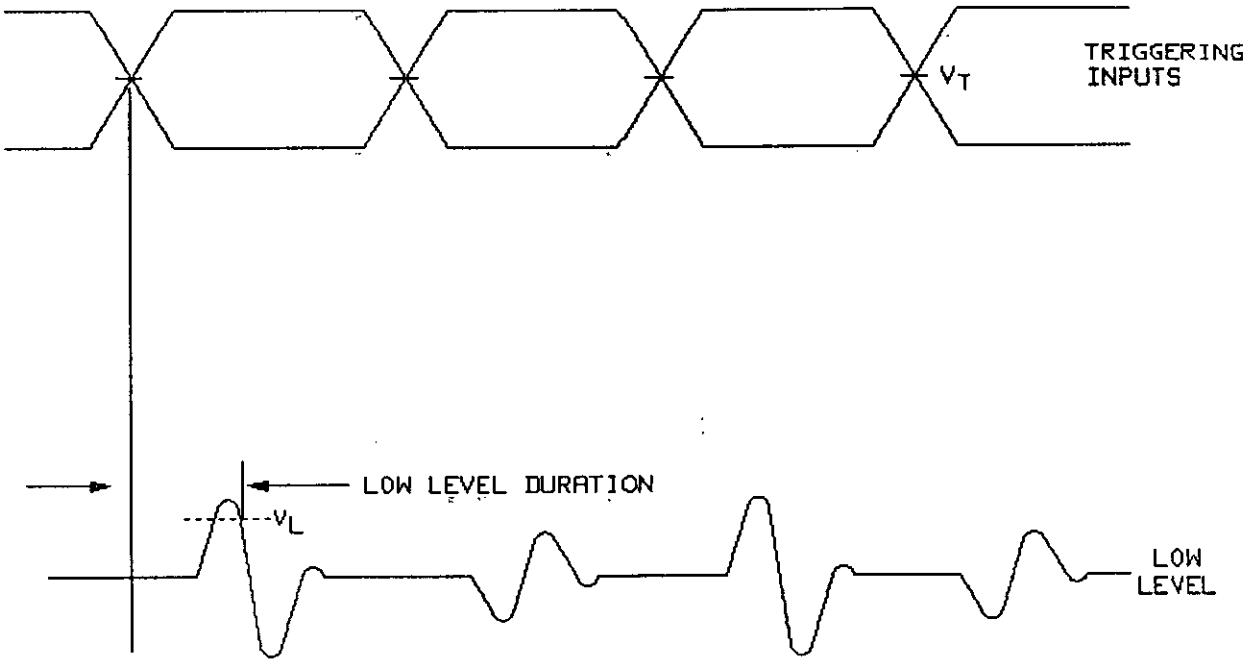
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NOTES: - Continued.

3. All outputs shall be loaded as specified on figure 8.

4. Abbreviations used are defined in 6.1.

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NOTES:

1. VT = 1.3 V, VL = 400 mV (ac coupled).

2. The crosstalk duration is measured from the triggering edge of the input to the point where output crosstalk returns to the specified voltage. If the crosstalk amplitude is smaller than the duration measurement voltage (VL), the crosstalk duration shall be recorded as 0 nS.

FIGURE 15. Output crosstalk, duration, test waveforms for table XXII.

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TABLE XXIII. Input capacitance test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
W1	DATA 0	94	2	35	pF	1-3
W2	DATA 1	44				
W3	DATA 2	42				
W4	DATA 3	87				
W5	DATA 4	38				
W6	DATA 5	88				
W7	DATA 6	39				
W8	DATA 7	89				
W9	DATA 8	41				
W10	DATA 9	91				
W11	DATA 10	92				
W12	DATA 11	43				
W13	DATA 12	93				
W14	DATA 13	45				
W15	DATA 14	95				
W16	DATA 15	46	2	35	pF	1-3
W17	DEC 0	54	2	20	pF	1-3
W18	DEC 1	4				
W19	DEC 2	53	2	20	pF	1-3
W20	-HALTI	14	2	25	pF	1-3

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TABLE XXIII. Input capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
W21	-RESETI	58	2	25	pF	1-3
W22	DECEN 1	2				
W23	-INPHASE	59				
W24	-ENINT	97				
W25	CLK	64	2	25	pF	1-3
W26	-BGACK	67	2	40	pF	1-3
W27	-BERR	17				
W28	-BR	20				
W29	-VI	62				
W30	-VPA	13				
W31	-DTACK	63	2	40	pF	1-3
W32	-DECEN 2	3	2	15	pF	1-3
W33	-DECEN 3	52	2	15	pF	1-3
W34	-IPL0	69	2	55	pF	1-3
W35	-IPL1	70				
W36	-IPL2	21	2	55	pF	1-3
W37	-INT1	48	2	30	pF	1-3
W38	-INT2	96				
W39	-INT3	47				
W40	-INT4	49	2	30	pF	1-3

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TABLE XXIII. Input capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
W41	-INT5	50	2	30	pF	1-3
W42	-INT6	100				
W43	-INT7	99	2	30	pF	1-3

NOTES:

1. Input capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance meter frequency	1 MHz	± 100 KHz	W1-W43
Capacitance meter voltage	15 mV	± 1.5 mV	W1-W43
Capacitance measurement	NA	± 1.65 pF	W1-W16
	NA	± 0.9 pF	W17-W19
	NA	± 1.15 pF	W20-W25
	NA	± 1.9 pF	W26-W31
	NA	± 0.65 pF	W32,W33
	NA	± 2.65 pF	W34-W36
	NA	± 1.4 pF	W37-W43

2. The capacitance meter used in this test shall be item C05. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XXIII.

3. Abbreviations used are defined in 6.1.

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TABLE XXIV. Output capacitance test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
X1	CLK	64	2	25	pF	1-3
X2	-VMA	71				
X3	-LDS	73				
X4	-UDS	24				
X5	ADDR1	26				
X6	ADDR2	77				
X7	ADDR3	29				
X8	ADDR4	28				
X9	ADDR5	27				
X10	ADDR6	78				
X11	ADDR7	30				
X12	ADDR8	32				
X13	ADDR9	80				
X14	ADDR10	82				
X15	ADDR11	83				
X16	ADDR12	84				
X17	ADDR13	35				
X18	ADDR14	36				
X19	ADDR15	85				
X20	ADDR16	34	2	25	pF	1-3

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TABLE XXIV. Output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
X21	ADDR17	33	2	25	pF	1-3
X22	ADDR18	81				
X23	ADDR19	79				
X24	ADDR20	76				
X25	ADDR21	74				
X26	ADDR22	86				
X27	ADDR23	37	2	25	pF	1-3
X28	FC0	65	2	40	pF	1-3
X29	FC1	16				
X30	FC2	22	2	40	pF	1-3
X31	READ	68	2	55	pF	1-3

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NOTES:

1. Output capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance meter frequency	1 MHz	± 100 KHz	X1-X31
Capacitance meter voltage	15 mV	± 1.5 mV	X1-X31
Capacitance measurement	NA NA NA	± 1.15 pF ± 1.9 pF ± 2.65 pF	X1-X27 X28-X30 X31

2. The capacitance meter used in this test shall be item C05. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XXIV.

3. Abbreviations used are defined in 6.1.

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TABLE XXV. Disable time test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Fig	Ref
Low to high impedance (tPLZ)							
Y1	CLK high to ADDR1	26	2.0	161.5	nS	16	1
Y2	CLK high to ADDR2	77					
Y3	CLK high to ADDR3	29					
Y4	CLK high to ADDR4	28					
Y5	CLK high to ADDR5	27					
Y6	CLK high to ADDR6	78					
Y7	CLK high to ADDR7	30					
Y8	CLK high to ADDR8	32					
Y9	CLK high to ADDR9	80					
Y10	CLK high to ADDR10	82					
Y11	CLK high to ADDR11	83					
Y12	CLK high to ADDR12	84					
Y13	CLK high to ADDR13	35	2.0	161.5	nS	16	1

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TABLE XXV. Disable time test - Continued.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Fig	Ref
Y14	CLK high to ADDR14	36	2.0	161.5	nS	16	1
Y15	CLK high to ADDR15	85					
Y16	CLK high to ADDR16	34					
Y17	CLK high to ADDR17	33					
Y18	CLK high to ADDR18	81					
Y19	CLK high to ADDR19	79					
Y20	CLK high to ADDR20	76					
Y21	CLK high to ADDR21	74					
Y22	CLK high to ADDR22	86					
Y23	CLK high to ADDR23	37	2.0	161.5	nS	16	1
High to high impedance (tPHZ)							
Y24	CLK high to ADDR1	26	2.0	246.0	nS	16	1
Y25	CLK high to ADDR2	77					
Y26	CLK high to ADDR3	29	2.0	246.0	nS	16	1

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TABLE XXV. Disable time test - Continued.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Fig	Ref
Y27	CLK high to ADDR4	28	2.0	246.0	nS	16	1
Y28	CLK high to ADDR5	27					
Y29	CLK high to ADDR6	78					
Y30	CLK high to ADDR7	30					
Y31	CLK high to ADDR8	32					
Y32	CLK high to ADDR9	80					
Y33	CLK high to ADDR10	82					
Y34	CLK high to ADDR11	83					
Y35	CLK high to ADDR12	84					
Y36	CLK high to ADDR13	35					
Y37	CLK high to ADDR14	36					
Y38	CLK high to ADDR15	85					
Y39	CLK high to ADDR16	34					
Y40	CLK high to ADDR17	33	2.0	246.0	nS	16	1

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TABLE XXV. Disable time test - Continued.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Fig	Ref
Y41	CLK high to ADDR18	81	2.0	246.0	nS	16	1
Y42	CLK high to ADDR19	79					
Y43	CLK high to ADDR20	76					
Y44	CLK high to ADDR21	74					
Y45	CLK high to ADDR22	86					
Y46	CLK high to ADDR23	37	2.0	246.0	nS	16	1
Y47	CLK high to -UDS	24	2.0	183.5	nS	16	2
Y48	CLK high to -LDS	73	2.0	183.5	nS	16	2
Low to high impedance(tPLZ)							
Y49	CLK high to READ	68	2.0	158.5	nS	16	3
Y50	CLK high (-BGACK low) to DATA 0	94	7.0	187.5	nS	16	4
Y51	CLK high (-BGACK low) to DATA 1	44					
Y52	CLK high (-BGACK low) to DATA 2	42	7.0	187.5	nS	16	4

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TABLE XXV. Disable time test - Continued.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Fig	Ref
Y53	CLK high (-BGACK low) to DATA 3	87	7.0	187.5	nS	16	4
Y54	CLK high (-BGACK low) to DATA 4	38					
Y55	CLK high (-BGACK low) to DATA 5	88					
Y56	CLK high (-BGACK low) to DATA 6	39					
Y57	CLK high (-BGACK low) to DATA 7	89					
Y58	CLK high (-BGACK low) to DATA 8	41					
Y59	CLK high (-BGACK low) to DATA 9	91					
Y60	CLK high (-BGACK low) to DATA 10	92					
Y61	CLK high (-BGACK low) to DATA 11	43					
Y62	CLK high (-BGACK low) to DATA 12	93	7.0	187.5	nS	16	4

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TABLE XXV. Disable time test - Continued.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Fig	Ref
Y63	CLK high (-BGACK low) to DATA 13	45	7.0	187.5	nS	16	4
Y64	CLK high (-BGACK low) to DATA 14	95					
Y65	CLK high (-BGACK low) to DATA 15	46	7.0	187.5	nS	16	4
High to high impedance (tPHZ)							
Y66	CLK high (-BGACK low) to DATA 0	94	7.0	187.5	nS	16	4
Y67	CLK high (-BGACK low) to DATA 1	44					
Y68	CLK high (-BGACK low) to DATA 2	42					
Y69	CLK high (-BGACK low) to DATA 3	87					
Y70	CLK high (-BGACK low) to DATA 4	38					
Y71	CLK high (-BGACK low) to DATA 5	88					
Y72	CLK high (-BGACK low) to DATA 6	39	7.0	187.5	nS	16	4

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TABLE XXV. Disable time test - Continued.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Fig	Ref
Y73	CLK high (-BGACK low) to DATA 7	89	7.0	187.5	nS	16	4
Y74	CLK high (-BGACK low) to DATA 8	41					
Y75	CLK high (-BGACK low) to DATA 9	91					
Y76	CLK high (-BGACK low) to DATA 10	92					
Y77	CLK high (-BGACK low) to DATA 11	43					
Y78	CLK high (-BGACK low) to DATA 12	93					
Y79	CLK high (-BGACK low) to DATA 13	45					
Y80	CLK high (-BGACK low) to DATA 14	95					
Y81	CLK high (-BGACK low) to DATA 15	46	7.0	187.5	nS	16	4

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NOTES:

1. Disable time test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 50 mV	Y1-Y81
Power supply current limit	1.09 A	NA	Y1-Y81
Pattern voltage high, (VH)	3.0 V	± 100 mV	Y1-Y81
low, (VL)	0.0 V	± 100 mV	Y1-Y81
Output voltage high, (VH)	NA	± 7.98 nS	Y1-Y23
	NA	± 12.20 nS	Y24-Y46
	NA	± 9.08 nS	Y47,Y48
	NA	± 7.83 nS	Y49
low, (VL)	NA	± 9.03 nS	Y50-Y81

2. The following procedure is required to perform this test.

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

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NOTES: - Continued.

3. Module inputs shall not violate the minimum input timing parameters referenced on figure 9.

4. Abbreviations used are defined in 6.1.

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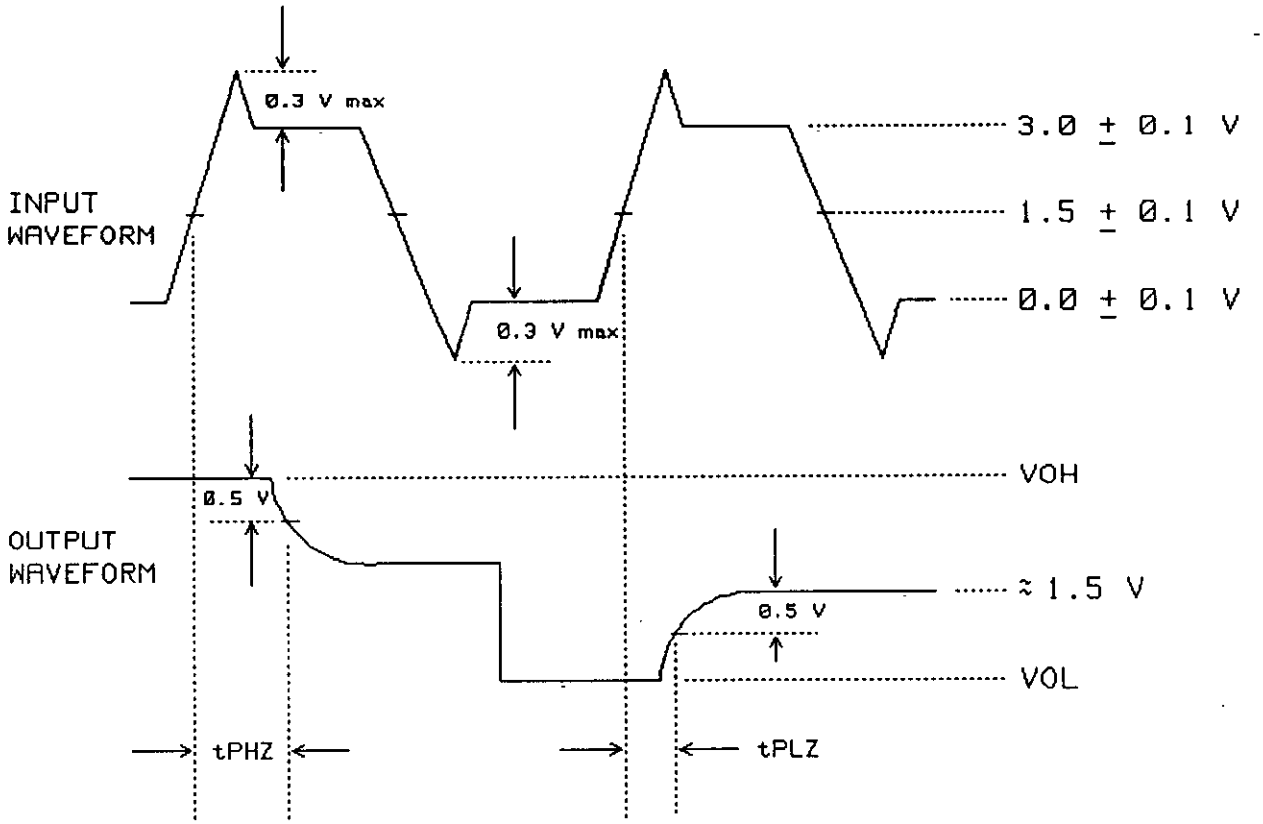


FIGURE 16a. Disable time waveforms for table XXV (timing measurement points).

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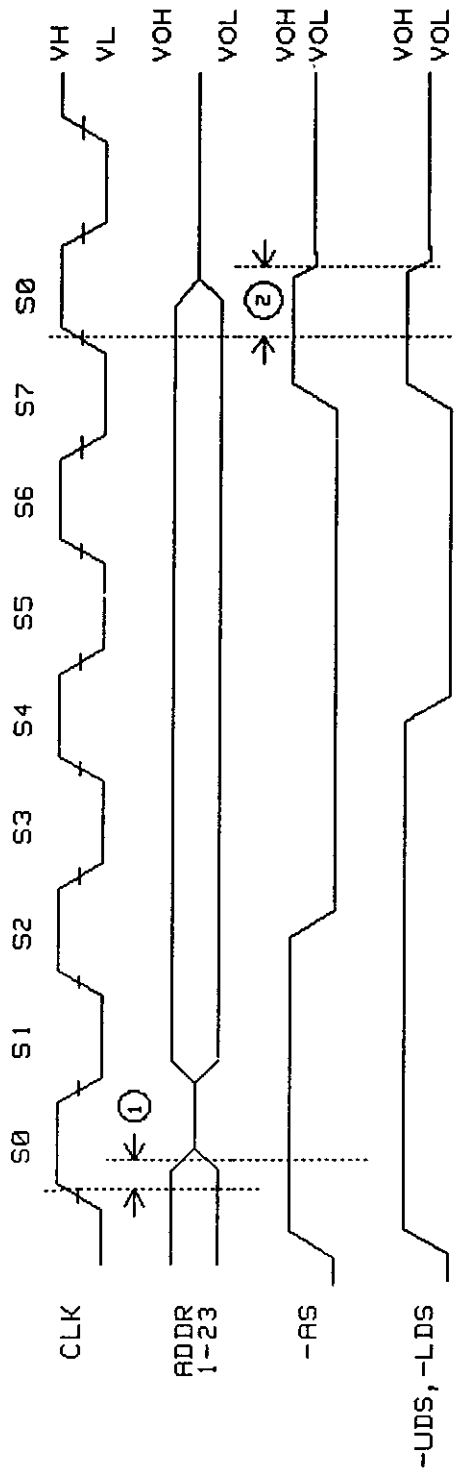


FIGURE 16b. Disable time test waveforms for table XXV.

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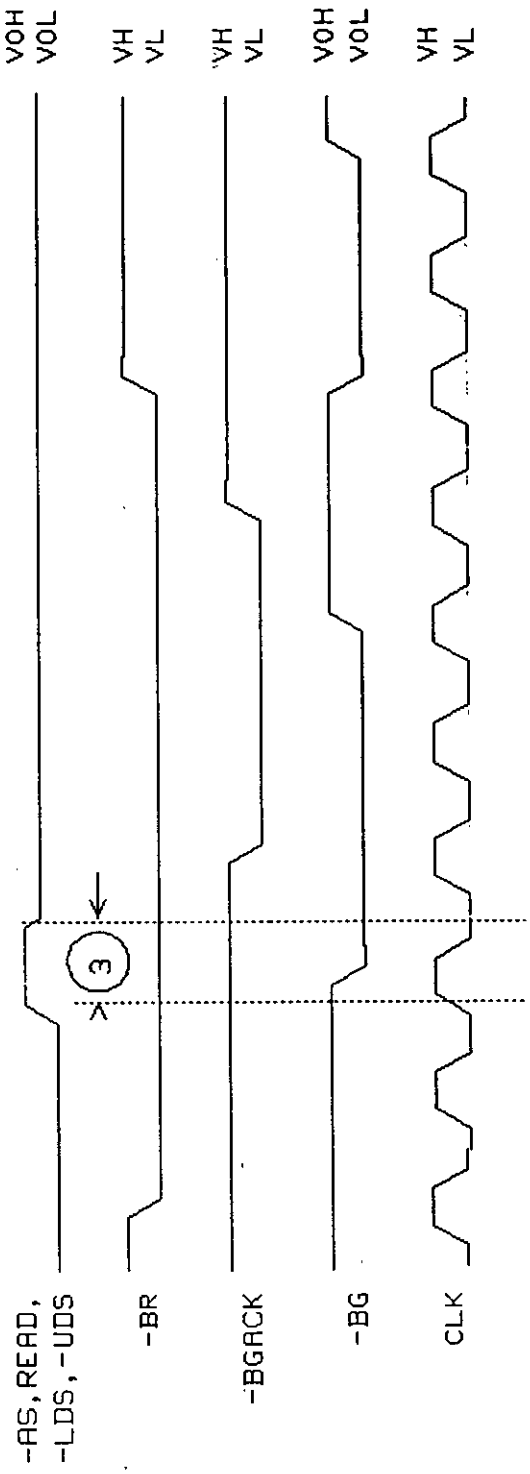


FIGURE 16c. Disable time test waveforms for table XXV.

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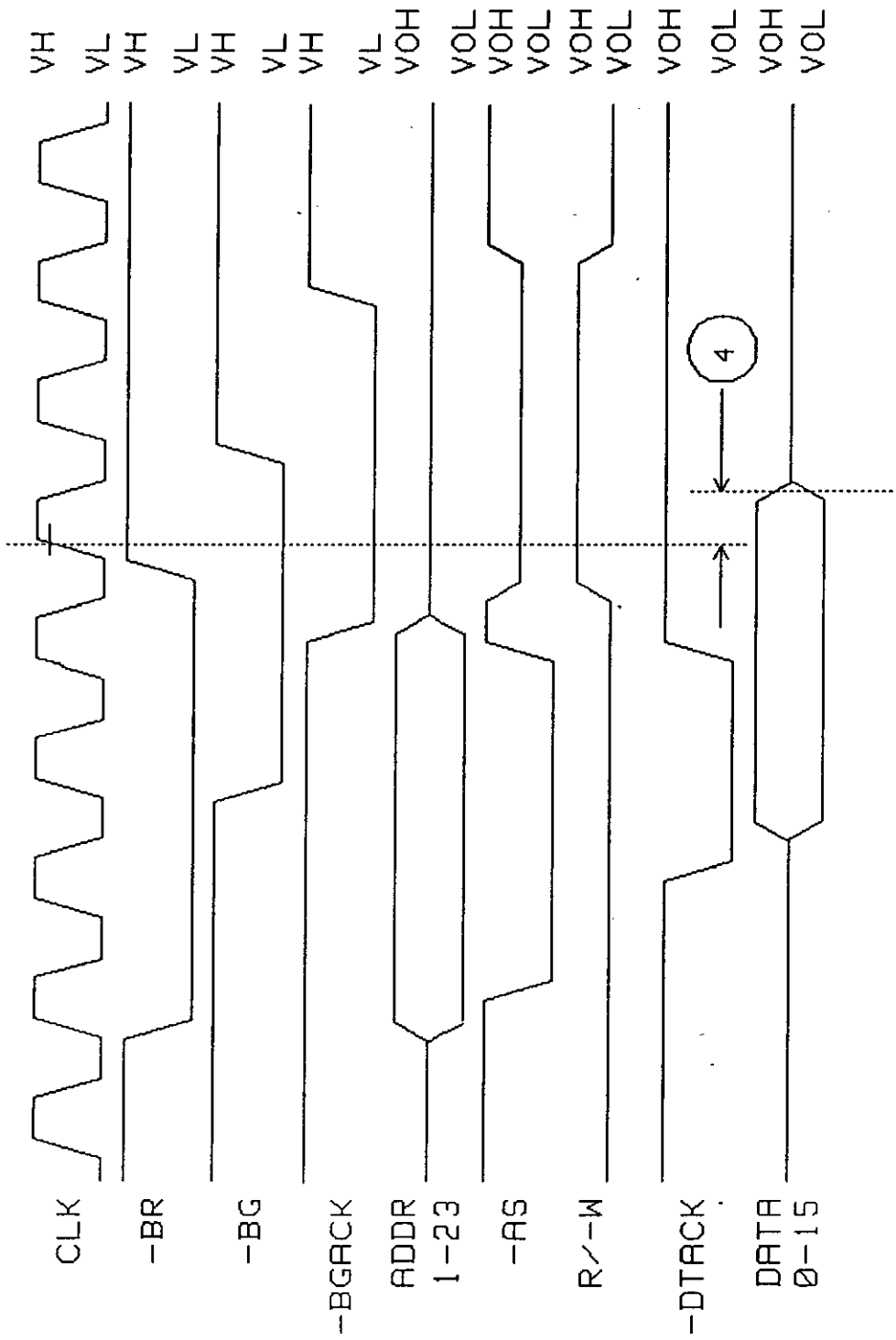


FIGURE 16d. Disable time test waveforms for table XXV.

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TABLE XXVI. Threshold test.

Test cond	Test description	Notes	Fig
Z	Threshold test	1-3	9

NOTES:

1. Threshold test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.75 V	± 47.5 mV	Z
Power supply current limit	1.09 A	NA	Z
Pattern voltage high, (VH) low, (VL)	 2.0 V 0.7 V	 ± 20.0 mV ± 7.0 mV	 Z Z
Output voltage high, (VH) low, (VL)	 NA NA	 ± 117.5 mV ± 25.0 mV	 Z Z

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test and verify the output patterns by monitoring the outputs and comparing the results with the output low and high levels specified. The module inputs shall not violate the minimum input timing requirements shown on figure 9.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

3. Abbreviations used are defined in 6.1.

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TABLE XXVII. Noise margin test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes	Fig
AA1	ADDR1	26	0	900	mV	1-4	17
AA2	ADDR2	77					
AA3	ADDR3	29					
AA4	ADDR4	28					
AA5	ADDR5	27					
AA6	ADDR6	78					
AA7	ADDR7	30					
AA8	ADDR8	32					
AA9	ADDR9	80					
AA10	ADDR10	82					
AA11	ADDR11	83					
AA12	ADDR12	84					
AA13	ADDR13	35					
AA14	ADDR14	36					
AA15	ADDR15	85					
AA16	ADDR16	34					
AA17	ADDR17	33					
AA18	ADDR18	81					
AA19	ADDR19	79					
AA20	ADDR20	76	0	900	mV	1-4	17

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TABLE XXVII. Noise margin test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes	Fig
AA21	ADDR21	74	0	900	mV	1-4	17
AA22	ADDR22	86	|	|	|	|	|
AA23	ADDR23	37	0	900	mV	1-4	17

NOTES:

1. Noise margin test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 0.25 V	AA1-AA23
Power supply current limit	1.09 A	NA	AA1-AA23
Noise margin high, (VH)	3.0 V	± 100 mV	AA1-AA23
low, (VL)	0.0 V	± 100 mV	AA1-AA23
Noise margin measurement	NA	± 45 mV	AA1-AA23

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) Use probe 'A' to verify the pattern voltage and pattern repetition rate of 1000 nS ± 200 nS, 50 percent duty cycle.

(d) For each output under test in table XXVII, connect probe 'B' and visually inspect the output for the most positive spike as indicated on figure 17.

(e) The maximum probe capacitance is 10 pF.

3. All outputs shall be loaded as specified on figure 8.

4. Abbreviations used are defined in 6.1.

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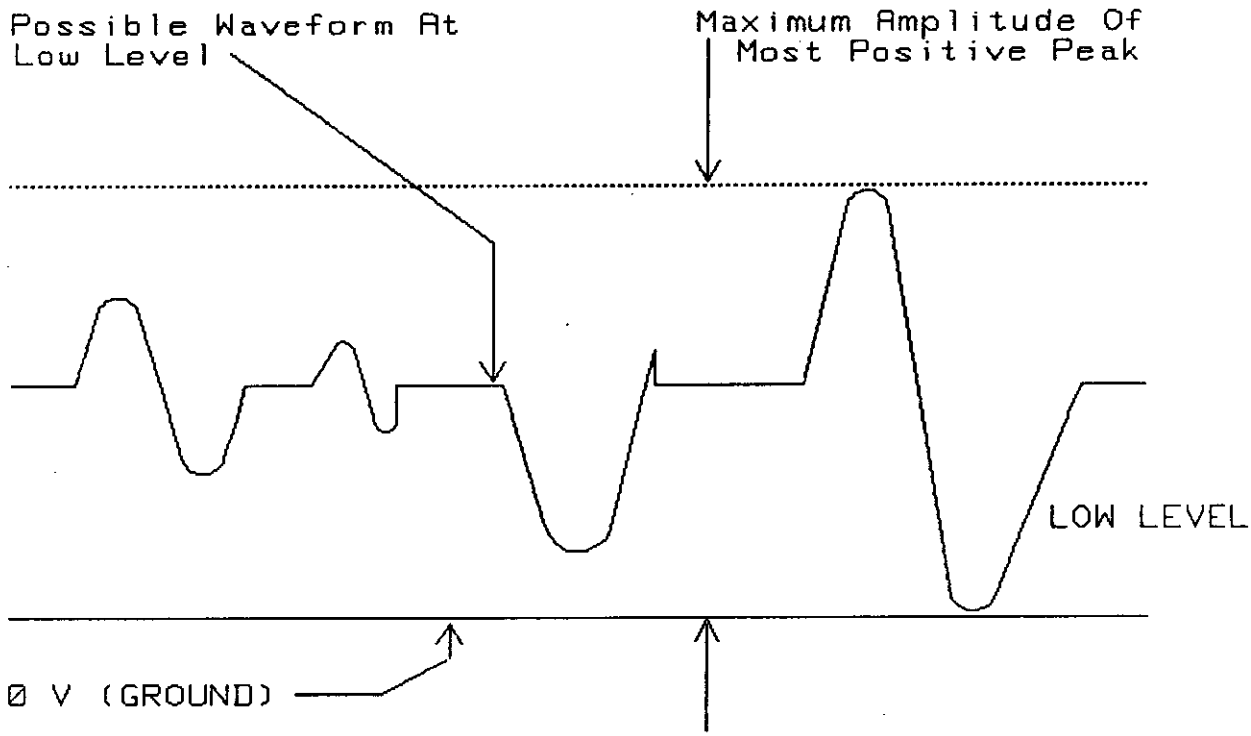


FIGURE 17. Noise margin test circuit for table XXVII.

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APPENDIX

MODULE FUNCTION

10 SCOPE

10.1 SCOPE. This appendix contains information regarding the module function. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance only.

20 APPLICABLE DOCUMENTS

20.1 This section is not applicable to this appendix.

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30 Module function. The NAH module contains a 68000 16 bit microprocessor with data bus transceivers to provide synchronous two-way communications between the microprocessor data bus and external data bus drive capability. With the exception of the specific NAH module implementation features described below, the NAH essentially contains a 68000 microprocessor that can be utilized to its full capability at a maximum clock frequency of 8.0 MHz. All 68000 instructions are supported by the NAH module. Reference Motorola's "MC68000 16 BIT MICROPROCESSOR, USER'S MANUAL" for the 68000 instruction set details.

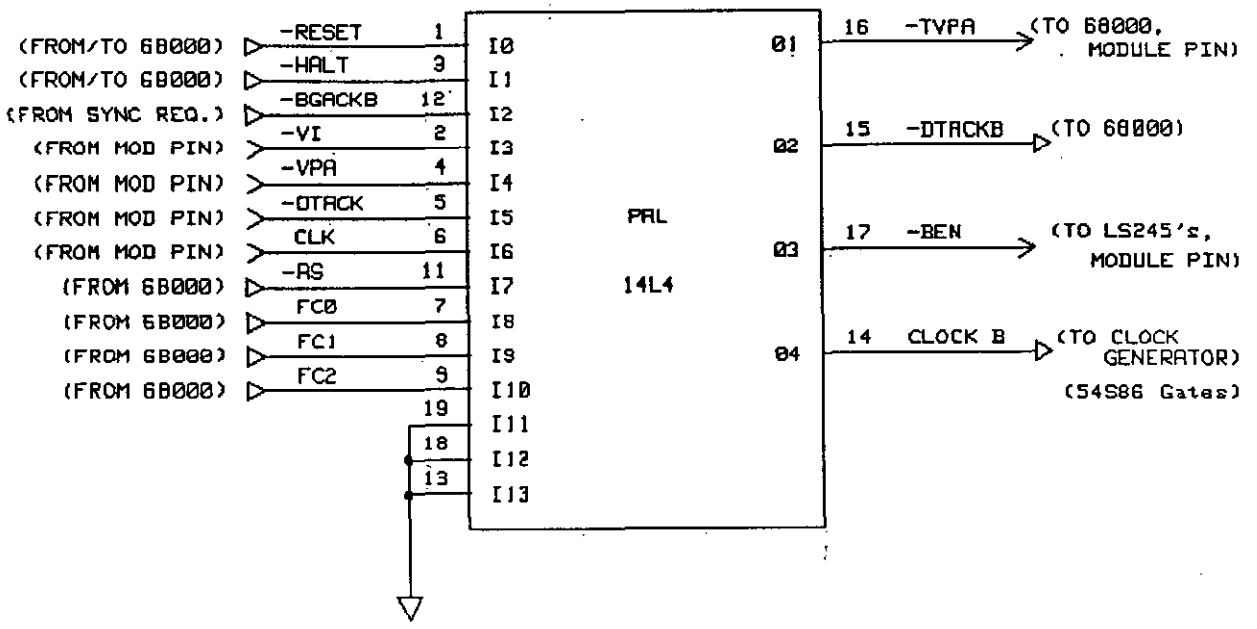
30.1 Clock synchronization. The NAH contains an internal register (54LS377) which is used to synchronize the -IPLO-2, -BR, -BERR and -BGACK inputs to the processor clock. With the exception of this synchronization, all other features of these inputs to the 68000 processor remain unchanged. An eight input priority encoder (54LS348) converts seven discrete interrupt input signals into a prioritized, 3 bit 68000 compatible code. The outputs of the priority encoder may be placed in the three-state mode (high impedance) which allows external logic to encode the interrupt inputs to the 68000.

30.2 Clock generation circuitry. Exclusive -OR gates are used in the clock generation circuitry and as buffers for system clock outputs (CLKB, -CLKB, PCLK). Open collector AND gates are used to implement a "two wire" -HALT and -RESET logic interface to the NAH. This allows for separate lines for the -HALT and -RESET inputs (-HALTI, -RESETI), and for the -HALT and -RESET outputs (-HALTO, -RESETO).

30.3 Processor status circuitry. The 3 to 8 bit decoder may be used to convert the 68000 function code outputs (FCO-2) to eight discrete signals indicating the status of the processor. The decoder is fully pinned out to allow its use for other applications.

30.4 Miscellaneous control decoding. A PAL is used to perform some miscellaneous control decoding on the module. The Boolean equations for the four PAL outputs -TVPA, -BEN, -DTACKB, and CLOCKB are illustrated in figure 1. For ease in determining the actual logic functions implemented in the PAL, a gate equivalent circuit for the PAL is illustrated in figure 2. Refer to table I for the active ("true") state assignments for the PAL inputs and outputs.

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PAL BOOLEANS:

$$-TVPA = \overline{-VPA + FC0 * FC1 * FC2 * \overline{-VI} * \overline{-AS}}$$

$$-BEN = (-HALT + \overline{-AS}) * -RESET * -BGACK$$

$$-DTACKB = -DTACK$$

$$CLOCK\ B = CLK$$

FIGURE 1. NAH module control decode PAL.

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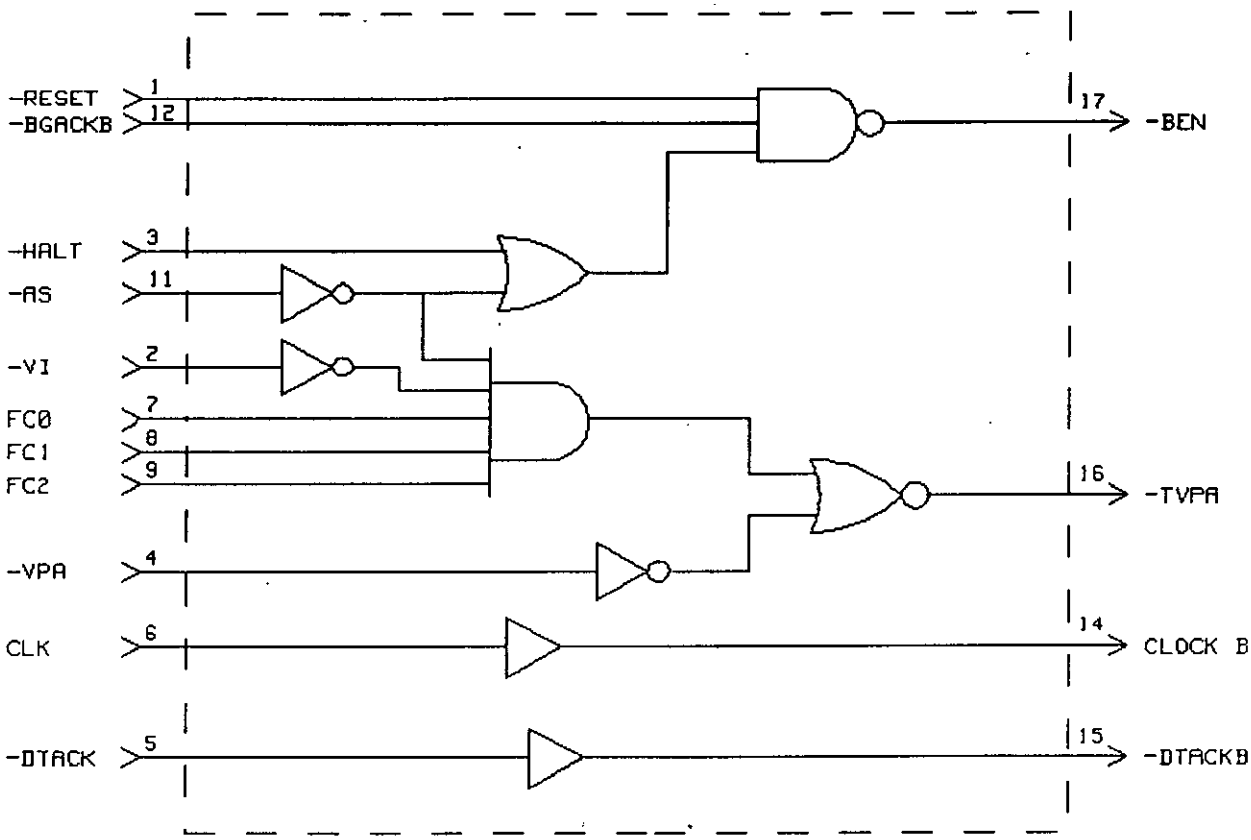


Figure 2. PAL14L4 internal logic diagram.

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TABLE I. Control decode PAL signal definition.

Electrical signal	Description	"True" Electrical state
-RESET	68000 Reset input/output	Low
-HALT	68000 Halt input/output	Low
-BGACKB	Buffered 68000 Bus Grant Acknowledge input (Output of Sync Register)	Low
-VI	Module automatic vector interrupt enable input	Low
-VPA	Module -VPA input	Low
-DTACK	Module -DTACK input	Low
CLK	Module clock input	High
-AS	68000 Address Strobe Output	Low
FC0	68000 Processor Status Output	High
FC1	68000 Processor Status Output	High
FC2	68000 Processor Status Output	High
-TVPA	68000 -VPA input and module test point	Low
-DTACKB	68000 Data Transfer Acknowledge input	Low
-BEN	Module data bus transceivers enable output	Low
CLOCK B	System clock generator output	High

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30.5 Interrupt synchronization precautions. The processor validates the code appearing on the -IPL0-2 lines by sampling the lines over two clock cycles, once during each instruction. If the code on the lines is the same for both of the samples, the interrupt is recognized and interrupt processing begins. If the codes are different, the processor continues executing the main program until two consecutive samples of the -IPL lines yield a valid interrupt. This scheme works quite well in discouraging spurious interrupts due to glitches on the -IPL0-2 lines. However, there are certain cases where a spurious interrupt may be recognized. For example, assume a level 1 interrupt is present on the interrupt input lines. Next, a level 4 interrupt arrives and is decoded by the priority encoder. During the output transition of the encoder, the 68000 may sample the -IPL0-2 lines. It is possible during the encoder transition for the -IPL0-2 lines to show a level 5 interrupt (going from level 1 to level 4). Next, during the next clock cycle, a level 7 interrupt arrives. The encoder will again go through a transition from level 4 to level 7. During this time, the encoder outputs can go through a level 5 interrupt again. If the processor samples the -IPL0-2 lines at this time, it will validate an erroneous level 5 interrupt. The processor will then vector incorrectly to the level 5 service routine. Circuits similar to the one shown on figure 3 will solve this problem. By latching the interrupt inputs into an external eight bit register and using a common clock signal (CLKB), the spurious level 5 interrupt in the example will never be seen by the processor. Note that the NAH module contains an internal register (54LS377) that synchronizes the -ILP0-2 inputs to the processor clock. All valid interrupts, however, will be acknowledged. Note that if the external interrupt inputs, -IPL0-2, are used directly (-ENINT input a HIGH), the proper encoded data must be applied to the NAH internal 54LS377 register with the required setup time specification satisfied.

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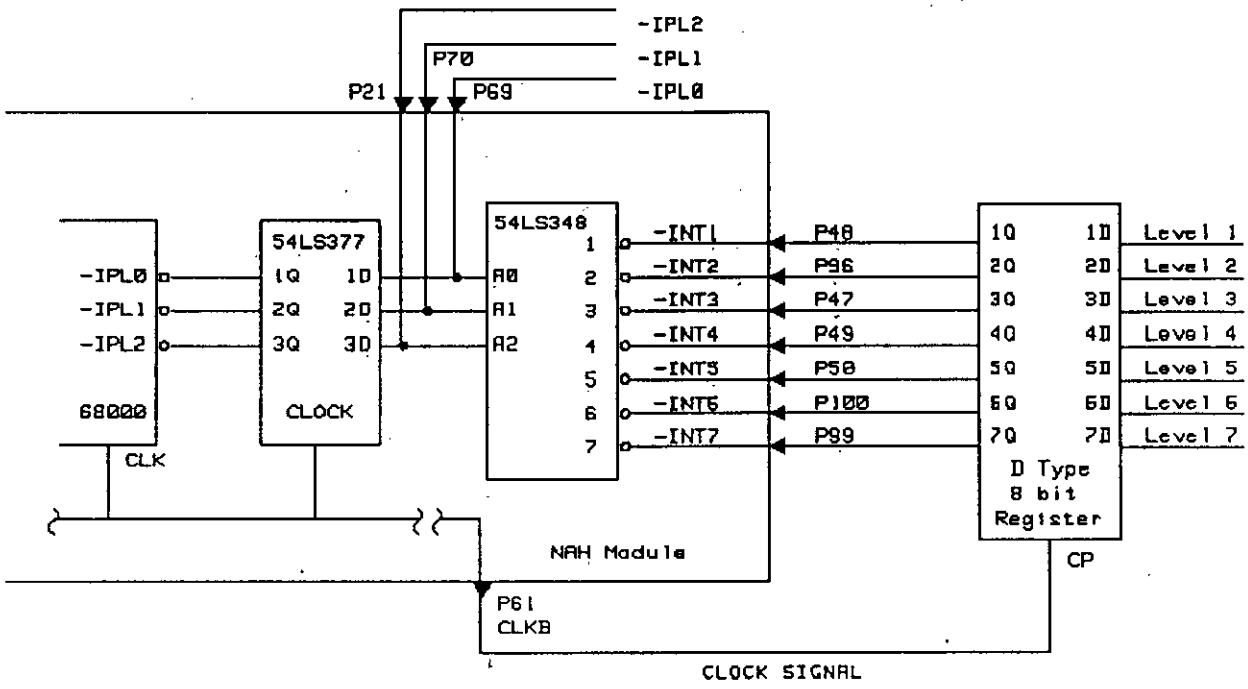


FIGURE 3. Interrupt synchronization circuit.

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30.6 Bus grant acknowledge (-BGACK) requirements. Bus grant acknowledge requirements (-BGACK input to the NAH cannot be asserted until the following conditions are met):

(a) A Bus Grant (-BG) has been received from the NAH.

(b) The NAH Address Strobe (-AS) output is inactive which indicates that the 68000 is not using the bus.

(c) The NAH Data Transfer Acknowledge (-DTACK) input is inactive which indicates that neither memory nor peripherals are using the bus.

(d) Bus Grant Acknowledge (-BGACK) is inactive which indicates that no other device is still claiming bus mastership. The -BGACK signal is usually configured as a wired-OR signal with the -BGACK signals from all potential bus masters connected together.

30.7 Crosstalk usage restrictions. To ensure crosstalk levels below 400mV on the -EN output, pin 98, allow a minimum of 50nS settling time with respect to the falling edge of the processor clock, CLKB, pin 61, during the beginning of state 3 of a write cycle.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-28787/353(SH)	Modules, Digital, Standard Electronic, Microprocessor, 68000 Module, Key Code NAH

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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